QUALIFICATION STANDARD AND PLAN (QSP)

Submarine Sonar Underwater Controller (Underwater Controller One)
NQual (AILE)

MITE ID
Residential Phase – 123877

Regular Force: SONAR Operator (00324)
Primary Reserve: (N/A)
Training Authority (TA): Commander Naval Training Group (CNTG)
Curriculum Control Authority (CCA): Naval Fleet School Atlantic (NFS(A)

MARPAC RDIMS #1387867
Chap 1 & 2 approved 01/12/2023
Chap 3 & 4 approved 18/01/2024

RECORD OF AMENDMENTS

1. The following versions of this QSP have been approved for use in support of the conduct of training. Amendments detailed in Annexes D and/or L.

Revision	Change Description	Date	Authority
1.0 (Mar 22)	QS Submission for Steady State Conversion from 3 Chap to 4 Chap QSP. CNPTG Changed to CNTG. NPTOR 4500-29 added.	1 Dec 23	NTG HQ
1.01	Steady state conversion of TP from 3 Chap to 4 Chap QSP format	18 Jan 24	CCA

NOTE:

Personnel programming courses must ensure that their hard copy of this document is current and identical to the electronic copy on the QSP SharePoint Site. See above for current changes.

TABLE OF CONTENTS

RECORD OF AMENDMENTS i
TABLE OF CONTENTS ii
QUALIFICATION STANDARD 1-0
CHAPTER 1 – QUALIFICATION STANDARD 1-1
AIM 1-1
GRANTING OF NQUAL 1-1
CO-QUALIFICATION 1-1
PREREQUISITES 1-1
CHAPTER 2 – PERFORMANCE OBJECTIVES 2-1
PO 001 – PLAN SONAR OPERATIONAL ACTIVITIES 2-1
PO 002 – LEAD SOUND ROOM IN SUPPORT OF OPERATIONS 2-4
PO 003 – DEVELOP SONAR TEAM IN SUPPORT OF OPERATIONS .. 2-7
CHAPTER 3 – TRAINING MANAGEMENT DETAILS 3-1
TRAINING STRATEGY 3-1
TRAINING DURATION 3-1
REQUIRED COURSE STAFFING 3-1
TRAINING LIMITATION 3-1
COURSE CAPACITY 3-1
DESIGNATED SCHEDULING AND COURSE LOADING 3-2
QUALIFICATION AND ASSESSMENT 3-2
UNSATISFACTORY PROGRESS 3-2
REMOVAL FROM TRAINING 3-2
RECERTIFICATION REQUIREMENTS 3-2
LANGUAGE OF INSTRUCTION 3-2
PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR) 3-3
POST COURSE REVIEWS (PCR) 3-3
COURSE REPORTING 3-3
CHAPTER 4 LESSON SPECIFICATIONS 4-1
PO 001 – PLAN SONAR OPERATIONAL ACTIVITIES 4-1
EO 001.01 PREPARE TACTICAL RECOMMENDATIONS FOR SONAR OPERATIONS **4-1**
EO 001.02 CONDUCT UWW BRIEF TO COMMAND **4-6**
PO 002 – LEAD SOUND ROOM IN SUPPORT OF OPERATIONS 4-9
EO 002.01 LEAD SOUND ROOM IN SUPPORT OF OPERATIONS **4-9**

EO 002.02 LEAD SOUND ROOM IN SUPPORT OF TOWED ARRAY OPERATIONS **4-13**

PO 003 – DEVELOP SONAR TEAM IN SUPPORT OF OPERATIONS 4-16

EO 003.01 DEVELOP SONAR TEAM IN SUPPORT OF OPERATIONS **4-16**

QUALIFICATION STANDARD ANNEXES 0

ANNEX A – MAIN REFERENCES 1

ANNEX B – TASK LIST 1

ANNEX C – QS SCALAR 1

ANNEX D – RECORD OF DECISIONS QSWB 1

RECORD OF DECISIONS WORKING GROUP **11**

RECORD OF AMENDMENT & AMPLIFICATION **12**

TRAINING PLAN ANNEXES 1

ANNEX E – RESOURCES, EQUIPMENT, SUPPLIES, AND ANCILLARIES 1

ANNEX F – COSTING INFORMATION 1

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS 1

ANNEX H – ASSESSMENT PLAN 1

ANNEX I – ASSESSMENT CHECKLISTS 1

ENABLING CHECK 001.01 PREPARE TACTICAL RECOMMENDATIONS **1**

ENABLING CHECK 001.02 CONDUCT UWW BRIEF TO COMMAND **4**

PERFORMANCE CHECK 001 PLAN SONAR OPERATIONAL ACTIVITIES **7**

ENABLING CHECK 002.01 LEAD SOUND ROOM IN SUPPORT OF OPERATIONS **10**

ENABLING CHECK 002.02 LEAD SOUND ROOM IN SUPPORT OF TOWED ARRAY OPERATIONS **12**

PERFORMANCE CHECK 002 LEAD SOUND ROOM IN SUPPORT OF OPERATIONS **15**

ENABLING CHECK 003.01 DEVELOP SONAR TEAM IN SUPPORT OF OPERATIONS **18**

ANNEX J – ON JOB PERFORMANCE RECORD (OJPR) 1

ANNEX K – TT PLAN 1

ANNEX L – RECORD OF DECISIONS TPWB 2

RECORD OF AMENDMENT & AMPLIFICATION **2**

ANNEX M – ACRONYMS AND ABBREVIATIONS 1

FOREWORD

1. This QS/TP is issued on the authority of the Commander of the Royal Canadian Navy and managed by the Commander Naval Training Group (CNTG) on behalf of the Assistant Chief of Naval Staff Personnel and Training (A/CNS P&T).

2. This QS/TP is based upon the Speciality Qualification, Underwater Controller One (UC1) published 26 Jul 2013, A-PD-055-002/PP-001;

3. This QS/TP publication is effective on receipt and supersedes Submarine Sonar Underwater Controller One QSP revision 2.2, dated 3 Feb 21. This QS/TP is designed to assist in developing, conducting, evaluating, and validating the Individual Training and Education (IT&E) required for training.

4. The QS at Chapters 1 and 2 were developed by the Qualification Standard Writing Board (QSWB) convened 21 Feb 22 at CFB Halifax working in collaboration with subject matter experts drawn from participating organizations. Updates to QS were reviewed and conducted 18-19 Aug 2023 at CFB Esquimalt. The subsequent TP at Chapters 3 and 4 were developed by the Training Plan Writing Board (TPWB) convened 21 Feb 22 at CFB Halifax working in collaboration with subject matter experts drawn from participating organizations. All aspects of this document were prepared in accordance with the concept of training outlined in A-P9-050 Series, Canadian Forces Manual of Individual Training and Education. Full details of the WBs are contained within Annex D and L.

5. The criteria for major and minor amendments, and the approval process, are identified in NPTORD 4500-22.

6. In circumstances where the QS and/or TP cannot be adhered to, submission of a Course Non-Concurrence, Deferrals, and Removals in accordance with NPTORD 4500-29 is required.

QUALIFICATION STANDARD

MARPAC RDIMS #1387867
Chap 1 & 2 approved 01/12/2023
Chap 3 & 4 approved 18/01/2024

CHAPTER 1 – QUALIFICATION STANDARD

AIM

1. The aim of this QSP is to establish the requirements and training for the Sonar Operator to perform their duties as a Submarine Sonar Underwater Controller (Underwater Controller One).

GRANTING OF NQUAL

2. A trainee will achieve the qualification AILE Submarine Sonar Underwater Controller (Underwater Controller One) when the standards listed in the Performance Objectives (POs) have been met.

CO-QUALIFICATION

3. Nil.

PREREQUISITES

4. To be selected for this course the loading authority must ensure that trainees meet the following prerequisites:

   a. Submarine Sonar Watch Leader (AKYG) qualified;

   b. Master Sailor Rank level;

   c. Sonar Operator Submarines (AILO) qualified; and

   d. Basic Submarine Course (AILS) qualified.

MARPAC RDIMS #1387867
Chap 1 & 2 approved 01/12/2023
Chap 3 & 4 approved 18/01/2024

CHAPTER 2 – PERFORMANCE OBJECTIVES

PO 001 – PLAN SONAR OPERATIONAL ACTIVITIES

1. Performance: Plan sonar operational activities.

2. Conditions:

   a. Given:

      (1) Assigned operational orders;

      (2) Operational Area;

      (3) Sonar equipment;

      (4) Ancillary equipment;

      (5) ORBAT (Order of Battle);

      (6) Access to request for information (RFI);

      (7) Access to unclass workstation;

      (8) Access to classified workstation;

      (9) Publications;

      (10) Immediate Command feedback; and

      (11) SOPs.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside/ashore: and

      (2) At sea.

3. Standard: The UC1 shall plan sonar operational activities to include:

   a. Preparing tactical recommendations for sonar operations to include:

      (1) Reviewing assigned activities to include but not limited to:

(a) Preparing UWW exercise support requests (D0045);

(b) Assisting with tests and trials with outside agencies (B0060);

(c) Preparing for operational missions; and

(d) Developing submarine special operations plans (C0205).

(2) Collecting data to support Command tactical decisions to include:

(a) Oceanographic data;

(b) Threat package;

(c) Pattern of life general intelligence;

(d) Providing data for RFIs (C0200) IAW A03.

i. Specify missing data; and

ii. Utilize RFI process.

(3) Analyzing data for impact on activities to include:

(a) Assessing UWW threat information (D0050);

(b) Identifying capacities and limitations of platforms (B0045); and;

(c) Interpreting oceanographic data (F0045).

(4) Recommending options for activities to include:

(a) Tactical movements of the submarine;

(b) Equipment usage for activities;

(c) Determining oceanographic effects on acoustic operations (C0150);

(d) Applying water space management (D0065) IAW Duty Plan;

(e) Equipment usage by extracting information from COMPLANS, EMCON Plan, and Frequency Plans (A0035) IAW A04;

(f) Calculating acoustic sensor range information (C0115);

(g) Calculating array tow speeds, tilt, and/or balancing (C0120);

(h) Developing noise monitoring plan (D0060) IAW A05 with recommendations for:

i. Noise rounds; and

ii. Sonar quiet states.

b. Conducting UWW brief to Command to include:

(1) Contributing to the master operations brief;

(2) Preparing UWW brief based on gathered information;

(3) Presenting UWW brief to Command to include:

(a) Displaying intelligence via UWW brief (B0035) to include;

i. Demonstrating confidence in material presented; and

ii. Utilizing professionalism.

(b) Providing recommendations so that the following occurs:

i. Aim is achieved;

ii. Submarine remains undetected; and

iii. Submarine remains safe.

(c) Providing justification for recommendations as requested.

4. References: Duty Plan, SOPs, A03, A04, and A05.

5. Task Numbers: A0035, B0035, B0045, B0060, C0115, C0120, C0150, C0200, C0205, D0045, D0050, D0060, D0065, and F0045.

6. Remarks: Nil.

PO 002 – LEAD SOUND ROOM IN SUPPORT OF OPERATIONS

1. Performance: Lead sound room in support of operations.

2. Conditions:

   a. Given:

      (1) Operational activity;

      (2) Operational area;

      (3) Sonar equipment;

      (4) Ancillary equipment;

      (5) ORBAT (Order of Battle);

      (6) Access to request for information (RFI);

      (7) Access to unclass workstation;

      (8) Access to classified workstation;

      (9) Immediate Command feedback;

      (10) Publications;

      (11) SOPs;

      (12) Submarine sound room.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside/ashore: and

      (2) At sea.

3. Standard: The UC1 shall lead sound room in support of operations to include:

   a. Supervising sonar operations in support of mission requirements to include:

      (1) Optimizing ship's acoustic signature (D0070) IAW A13 to include:

(a) Conducting noise rounds (C0130);

(b) Collecting and reporting Vibration Analysis (VA) for watch data;

(c) Maintaining quiet running bill (D0055); and

(d) Making recommendations for movements to optimize tactical posture.

(2) Validating sensor data to develop the tactical picture to include:

(a) Directing the correlation of active / passive / non-acoustic information (C0105);

(b) Performing Passive Aural Listening (PAL) (C0195);

(c) Localizing passive acoustic contacts (C0085);

(d) Tracking acoustic data (C0070);

(e) Classifying passive acoustic data (C0080);

(f) Amplifying track information (E0020);

(g) Analyzing passive acoustic data (C0065); and

(h) Evaluating correlated acoustic information (F0005).

b. Directing towed array system operations (C0155) IAW A21, A06, and SOPs;

c. Providing Command with Situation, Assessment, Recommendation (SAR) reports IAW A02 and SOPs;

d. Contributing to intelligence gathering operations (B0030);

e. Monitoring operation of all submarine sonar equipment (C0235) IAW A10 to include:

(1) Ensuring efficient running of the noise detection organization IAW A13;

(2) Ensuring the correct operation of all sonar equipment IAW A6; and

(3) Communicating to Command of any sonar equipment defects.

f. Directing the maintenance of operational logs (B0055) IAW A14 to include:

(1) Recording acoustic intelligence data (F0010);

(2) Manually written logs IAW SOPs;

(3) Digital logs IAW SOPs; and

(4) Maintaining sonar stateboard (B0015).

4. References: SOPs, A02, A06, A10, A13, A14, and A21.

5. Task Numbers: B0015, B0030, B0055, C0065, C0070, C0080, C0085, C0105, C0130, C0155, C0235, C0195, D0055, D0070, E0020, F0005, F0010.

6. Remarks: Nil.

PO 003 – DEVELOP SONAR TEAM IN SUPPORT OF OPERATIONS

1. Performance: Develop sonar team in support of operations.

2. Conditions:

   a. Given:

      (1) Operational activity;

      (2) Training activity (refresher, familiarization, CRRs);

      (3) Operational area;

      (4) Sonar equipment;

      (5) Ancillary equipment;

      (6) ORBAT (Order of Battle);

      (7) Access to request for information (RFI);

      (8) Access to unclass workstation;

      (9) Access to classified workstation;

      (10) Immediate Command feedback;

      (11) Publications;

      (12) SOPs;

      (13) Submarine sound room.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside/ashore: and

      (2) At sea.

3. Standard: The UC1 shall develop sonar team in support of operations to include:

   a. Supporting Command with individual unit/ships packages to include:

(1) Conducting Unit On the Job Training (OJT), Ship's Development Package (SDP) and Pre-Information Package (PIP) (G0085);

(2) Compiling/distributing Unit OJT, SDP, PIP reports (G0040) to include, but not limited to:

(a) Submarine safety IAW A10;

(b) Basic Submarine Qualification; and

(c) Submarine Sonar Conversions.

(3) Compliance monitoring; and

(4) Compliance enforcement.

b. Planning/designing training exercises (G0010) to include:

(1) Inputting training scenarios (G0090);

(2) CRRs; and

(3) Recommending improvements to training (L0035).

c. Conducting continuation/proficiency training (G0080);

d. Conducting team training events (G0065) to include;

(1) Assessing team training performance (G0120);

(2) Correcting team training deficiencies (G0070); and

(3) Providing section/unit with feedback (G0140).

e. Conducting individual training events (G0050) to include;

(1) Assessing individual training performance (G0115); and

(2) Correcting individual training deficiencies (G0055).

4. References: A6, A10, A13, A14, and A21.

5. Task Numbers: G0010, G0040, G0050, G0055, G0065, G0070, G0080, G0085, G0090, G0115, G0120, G0140, and L0035.

6. Remarks: UC1 shall be formatively assessed throughout the training plan for this Performance Objective. There will not be a formal summative assessment for all items listed in the standard.

TRAINING PLAN

MARPAC RDIMS #1387867
Chap 1 & 2 approved 01/12/2023
Chap 3 & 4 approved 18/01/2024

CHAPTER 3 – TRAINING MANAGEMENT DETAILS

TRAINING STRATEGY

1. The residential training strategy was selected for this course.

TRAINING DURATION

2. Based on existing resources and approved instructional design requirements, the approved Course Design authorizes 21 training days.

REQUIRED COURSE STAFFING

3. The following key personnel are required to conduct the training.

Instructors				
Rank	MOSID	Qualifications Required	Qty	PO / EO
Min PO2	00324	AILE	1	All

Guest Lecturers/Others (May be civilian)				
Lecturer	# Instructor	Subject	EO #	# period
NFS(A) Sub Div	1	Develop Sonar Team	003.01	1
Trinity / METOC	1	Intelligence gathering	003.01	1
CFMWC	1	Resources and supports available	003.01	1
MOC Advisor	1	Leadership, roles, and responsibilities	003.01	1

TRAINING LIMITATION

4. A training limitation may be due to constraints of resources, personnel or equipment. A training limitation is the difference between what a qualified individual must be able to do and what training can prepare them to do. The following limitations are identified below: nil.

COURSE CAPACITY

5. The capacity per session is based on the guidelines in CFITES Vol 5 – Instructor to Learner Ratios:

a. Optimum – 4;

b. Maximum Load – 6; and

c. Minimum – 2

MARPAC RDIMS #1387867
Chap 1 & 2 approved 01/12/2023
Chap 3 & 4 approved 18/01/2024

DESIGNATED SCHEDULING AND COURSE LOADING

6. Training will be scheduled utilizing the HRMS Training (MITE) module to meet the training requirements (IAW NPTORD 4540-2).

7. The authority to load members onto this course will be IAW NPTORD 4540-43. NPTG HQ is to be included as an info addressee on all loading messages. Course loading will be accomplished by the Career Manager.

QUALIFICATION AND ASSESSMENT

8. This is a pass/fail course. Students must successfully complete all POs to achieve the qualification.

9. Students will be monitored and assessed both on performance and conduct throughout the course. Counselling, progress monitoring, and maintenance of student files will be in accordance with NPTORD 4500. Although not mandated specifically in the POs, the general demands of service members and the leadership demands of junior and senior leaders form an integral part of individual training. As such, students are presumed to possess the leader and/or follower attributes commensurate with their rank.

UNSATISFACTORY PROGRESS

10. Any student who fails to successfully complete all POs in the allocated number of attempts will be referred to the Commandant (Cmdt) or Commanding Officer (CO) of the training establishment (TE) to determine the requirement for a TRB IAW NPTORD 4500-54.

REMOVAL FROM TRAINING

11. A TE CO/Cmdt may direct that a student be removed from training for reasons of deficiency in performance, conduct (discipline, attitude, teamwork, safety, leadership), missed training, medical, and other non-performance/conduct related issues. In these cases, Cease Training Procedures are to be strictly adhered to IAW NPTORD 4500-29 or 4500-54.

RECERTIFICATION REQUIREMENTS

12. There are no additional requirements to maintain currency in this qualification.

LANGUAGE OF INSTRUCTION

13. The language of instruction shall be in accordance with the Delivery of Training and Education in Both Official Languages (DAOD 5039-6).

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

14. PLARs shall be conducted in accordance with DAOD 5031-1 Canadian Forces Military Equivalencies Program, through the RCN PLAR submissions coordinator.

POST COURSE REVIEWS (PCR)

15. PCRs are to be conducted on all steady state courses IAW NPTORD 4500-23.

COURSE REPORTING

16. Formal Course: A Canadian Forces Course Report, Form CF 377 shall be prepared and distributed in accordance with DAOD 5031-9. The Course Report shall be descriptive, identifying the trainee's strengths and weaknesses with specific comments on any PO for which the trainee demonstrated outstanding performance or had difficulty achieving the standard, any conduct relevant to course performance, and any conditions or situations that affected course performance.

CHAPTER 4 LESSON SPECIFICATIONS

PO 001 – PLAN SONAR OPERATIONAL ACTIVITIES

EO 001.01 PREPARE TACTICAL RECOMMENDATIONS FOR SONAR OPERATIONS

1. Performance: Prepare tactical recommendations for sonar operations.

2. Conditions:

   a. Given:

      (1) Assigned operational orders;

      (2) Operational Area;

      (3) ORBAT (Order of Battle);

      (4) DWAN;

      (5) CSNI; and

      (6) References and publications;

   b. Denied: time in excess of deadlines;

   c. Environment: Classroom.

3. Standard: The trainee shall prepare tactical recommendations for sonar operations to include:

   a. Reviewing assigned activities to include but not limited to:

      (1) Preparing UWW exercise support requests (D0045);

      (2) Assisting with tests and trials with outside agencies (B0060);

      (3) Preparing for operational missions; and

      (4) Developing submarine special operations plans (C0205);

   b. Collecting data to support Command tactical decisions to include:

      (1) Oceanographic data;

      (2) Threat package; and

      (3) Pattern of life general intelligence;

MARPAC RDIMS #1387867
Chap 1 & 2 approved 01/12/2023
Chap 3 & 4 approved 18/01/2024

(4) Providing data for RFIs (C0200) IAW A03 to include:

(a) Specify missing data; and

(b) Utilize RFI process;

c. Analyzing data for impact on activities to include:

(1) Assessing UWW threat information (D0050);

(2) Identifying capacities and limitations of platforms (B0045); and;

(3) Interpreting oceanographic data (F0045);

d. Recommending options for activities to include:

(1) Tactical movements of the submarine to include;

(a) Ranges for detection / counter-detection; and

(b) Movements based on ranges.

(c) Depth brackets;

(d) Rudder and hydroplane movements; and

(e) Speeds;

(2) Equipment usage for activities;

(3) Determining oceanographic effects on acoustic operations (C0150);

(4) Applying water space management (D0065) IAW Duty Plan;

(5) Equipment usage by extracting information from COMPLANS, EMCON Plan, and Frequency Plans (A0035) IAW A04;

(6) Calculating acoustic sensor range information (C0115);

(7) Calculating array tow speeds, tilt, and/or balancing (C0120);

(8) Developing noise monitoring plan (D0060) IAW A05 with recommendations for:

(a) Noise rounds; and

(b) Sonar quiet states.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
1. Identify planning requirements based on assigned activities.	Method: Interactive Lecture Media: Instructor Environment: SECRET Classroom	1 x 50 min	A1, A6, A7, A8, A9
2. Collect required data to include: - established appropriate planning timelines and priorities - identify required data - source and collect available data	Method: Research activity Media: Scenario Environment: SECRET Classroom	1 x 50 min	A1, A6, A7, A8, A9
3. Complete an RFI			
4. Discuss tactical recommendations in sonar operational planning	Method: Interactive lecture Media: Instructor Environment: SECRET classroom	2 x 50 min	A1, A6, A7, A8, A9

Teaching Points	Instructional Strategies	Time	References/Remarks
5. Prepare tactical recommendations based on available information to include: - -conduct data analysis - identify oceanographic considerations affecting tactics - Assessing UWW threat information - Capacities and limitations of platforms; - Interpretation of oceanographic data - Considerations and options for movements and activities - Water space management - Equipment usage and restrictions - Ranges for detection / counter-detection - Towed array speeds and movement - Noise monitoring plan (noise rounds and sonar quiet states) - Depth brackets - Rudder and planing movements - Speeds	Method: Guided Practice Media: Scenario Environment: SECRET classroom	7 x 50 min	A1, A6, A7, A8, A9

5. Total EO Time: Total: 25 x 50 minute periods as follows:

   a. 11 x 50 min period; and

   b. 14 x 50 min periods – EC (total time).

6. Training and Learning Aids: Scenario (operational mission to be assigned).

7. Remarks: nil.

EO 001.02 CONDUCT UWW BRIEF TO COMMAND

1. Performance: Conduct UWW brief to Command.

2. Conditions:

    a. Given:

        (1) Assigned operational orders;

        (2) Operational Area;

        (3) Data collected in EO 001.01;

        (4) ORBAT (Order of Battle);

        (5) DWAN;

        (6) CSNI;

        (7) References and publications; and

        (8) Immediate Command feedback;

    b. Denied: nil;

    c. Environment: classroom.

3. Standard: The trainee shall conduct UWW brief to Command by:

    a. Contributing to the master operations brief;

    b. Preparing UWW brief based on gathered information to include:

        (1) Introduction;

        (2) Classification level;

        (3) References;

        (4) Oceanography;

        (5) Threat platforms; and

        (6) Recommendations;

    c. Presenting UWW brief to Command to include:

        (1) Displaying intelligence via UWW brief (B0035) to include;

MARPAC RDIMS #1387867
Chap 1 & 2 approved 01/12/2023
Chap 3 & 4 approved 18/01/2024

(a) Demonstrating confidence in material presented to include:

i. Maintaining appropriate eye contact;

ii. Employing appropriate body language and posture;

iii. Maintaining a confident, calm presence; and

iv. Displaying evidence of preparation and in-depth knowledge of material being presented;

(b) Utilizing professionalism to include:

i. Using respectful tone, language, and delivery; and

ii. Speaking in clear, concise, and brief manner;

(2) Providing recommendations so that the following occurs:

(a) Aim is achieved;

(b) Submarine remains undetected; and

(c) Submarine remains safe;

(3) Providing justification for recommendations as requested, to include:

(a) Explaining reasoning for recommendations;

(b) Answering questions with facts;

(c) Responding to questions with appropriate tone, language, and delivery; and

(d) Displaying critical thinking to Command queries to include:

i. Applying logical thinking to questions posed; and

ii. Replying to queries based on available data and situational awareness.

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
1. Conduct an UWW Brief	Method: EDI Media: Instructor Environment: SECRET Classroom	2 x 50 min	
2. Practice UWW briefing	Method: Practice Media: PPT Environment: SECRET Classroom	1 x 50 min	

5. Total EO Time: Total: x 50 minute periods as follows:

   a. 3 x 50 min period; and

   b. 7 x 50 min periods – EC (total time).

6. Training and Learning Aids: Scenario (operational mission to be assigned).

7. Remarks: nil.

PO 002 – LEAD SOUND ROOM IN SUPPORT OF OPERATIONS

EO 002.01 LEAD SOUND ROOM IN SUPPORT OF OPERATIONS

1. Performance: Lead sound room in support of operations.

2. Conditions:

    a. Given:

        (1) Operational activity;

        (2) Operational area;

        (3) Sonar equipment;

        (4) Ancillary equipment;

        (5) ORBAT (Order of Battle);

        (6) DWAN;

        (7) CSNI;

        (8) Supervision (indirect);

        (9) Immediate Command feedback;

        (10) Sound room personnel; and

        (11) References and publications;

    b. Denied: Assistance in performance of job duties;

    c. Environment: Submarine trainer (VCSCTT or TWS).

3. Standard: The trainee shall lead sound room in support of operations to include:

    a. Supervising sonar operations in support of mission requirements to include:

        (1) Optimizing ship's acoustic signature (D0070) IAW A13 to include:

            (a) Conducting noise rounds (C0130);

            (b) Collecting and reporting Vibration Analysis (VA) for watch data;

MARPAC RDIMS #1387867
Chap 1 & 2 approved 01/12/2023
Chap 3 & 4 approved 18/01/2024

(c) Maintaining quiet running bill (D0055); and

(d) Making recommendations for movements to optimize tactical posture;

(2) Validating sensor data to develop the tactical picture to include:

(a) Directing the correlation of active / passive / non-acoustic information (C0105);

(b) Performing Passive Aural Listening (PAL) (C0195);

(c) Localizing passive acoustic contacts (C0085);

(d) Tracking acoustic data (C0070);

(e) Classifying passive acoustic data (C0080);

(f) Amplifying track information (E0020);

(g) Analyzing passive acoustic data (C0065); and

(h) Evaluating correlated acoustic information (F0005);

b. Providing Command with Situation, Assessment, Recommendation (SAR) reports IAW A02 and SOPs;

c. Contributing to intelligence gathering operations (B0030);

d. Monitoring operation of all submarine sonar equipment (C0235) IAW A10 to include:

(1) Ensuring efficient running of the noise detection organization IAW A13;

(2) Ensuring the correct operation of all sonar equipment IAW A6;

(3) Communicating to Command of any sonar equipment defects; and

(4) Ensuring routine maintenance and cleanliness of the Control Room;

e. Directing the maintenance of operational logs (B0055) IAW A14 to include:

(1) Recording acoustic intelligence data (F0010);

(2) Manually written logs IAW SOPs;

(3) Digital logs IAW SOPs; and

(4) Maintaining sonar stateboard (B0015).

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
1. Discuss Signature Reduction Organization (SRO).	Method: Interactive Lecture Media: Instructor Environment: SECRET Classroom	1 x 50 min	A13
1. Maintain operational logs	Method: Interactive lecture Media: Instructor Environment: SECRET classroom	1 x 50 min	A6, A14
3. Conduct roles and responsibilities of UC1. - Validate tactical picture - Assess situation - Communicate tactical picture - Run the noise detection organization - Direct record keeping - Provide Command with SAR reports - Lead Sound Room	Method: EDI Media: Instructor Environment: Trainer (VCSCTT or TWS)	5 x 50 min	A6, A10, A19
4. Lead sound room in support of operations. - Validate tactical picture - Assess situation - Communicate tactical picture - Run the noise detection organization - Direct record keeping - Provide Command with SAR reports - Lead Sound Room	Method: Practice Media: Real Equipment Environment: Trainer (VCSCTT or TWS)	24 x 50 min	A6, A10, A19

5. Total EO Time: Total: 38 x 50 minute periods as follows:

a. 31 x 50 min period; and

b. 7 x 50 min periods – EC (total time).

6. Training and Learning Aids: Scenario (operational mission to be assigned).

7. Remarks: nil.

EO 002.02 LEAD SOUND ROOM IN SUPPORT OF TOWED ARRAY OPERATIONS

1. Performance: Lead sound room in support of towed operations.

2. Conditions:

    a. Given:

        (1) Towed array operational activity;

        (2) Operational area;

        (3) Sonar equipment;

        (4) Ancillary equipment;

        (5) ORBAT (Order of Battle);

        (6) DWAN;

        (7) CSNI;

        (8) Supervision (indirect);

        (9) Immediate Command feedback;

        (10) Sound room personnel; and

        (11) References and publications;

    b. Denied: Assistance in performance of job duties;

    c. Environment: Submarine trainer (VCSCTT or TWS).

3. Standard: The trainee shall lead sound room in support of operations to include:

    a. Supervising sonar operations in support of mission requirements to include:

        (1) Optimizing ship's acoustic signature (D0070) IAW A13 to include:

            (a) Conducting noise rounds (C0130);

            (b) Collecting and reporting Vibration Analysis (VA) for watch data;

            (c) Maintaining quiet running bill (D0055); and

(d) Making recommendations for movements to optimize tactical posture;

(2) Validating sensor data to develop the tactical picture to include:

(a) Directing the correlation of active / passive / non-acoustic information (C0105);

(b) Performing Passive Aural Listening (PAL) (C0195);

(c) Localizing passive acoustic contacts (C0085);

(d) Tracking acoustic data (C0070);

(e) Classifying passive acoustic data (C0080);

(f) Amplifying track information (E0020);

(g) Analyzing passive acoustic data (C0065); and

(h) Evaluating correlated acoustic information (F0005);

b. Providing Command with Situation, Assessment, Recommendation (SAR) reports IAW A02 and SOPs;

c. Contributing to intelligence gathering operations (B0030);

d. Monitoring operation of all submarine sonar equipment (C0235) IAW A10 to include:

(1) Ensuring efficient running of the noise detection organization IAW A13;

(2) Ensuring the correct operation of all sonar equipment IAW A6;

(3) Communicating to Command of any sonar equipment defects; and

(4) Ensuring routine maintenance and cleanliness of the Control Room;

e. Directing the maintenance of operational logs (B0055) IAW A14 to include:

(1) Recording acoustic intelligence data (F0010);

(2) Manually written logs IAW SOPs;

(3) Digital logs IAW SOPs; and

(4) Maintaining sonar stateboard (B0015).

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
1. Discuss towed array operations. - direct towed array operations - deploy towed array	Method: Interactive Lecture Media: Instructor Environment: SECRET Classroom	1 x 50 min	A20, A21, B2
2. Conduct duties and responsibilities of UC1 in support of towed array operations	Method: EDI Media: Instructor Environment: Trainer (VCSCTT or TWS)	5 x 50 min	A20, A21, B2
3. Lead sound room in support of towed array operations	Method: Practice Media: Real Equipment Environment: Trainer (VCSCTT or TWS)	23 x 50 min	A20, A21, B2

5. Total EO Time: Total: 36 x 50 minute periods as follows:

   a. 29 x 50 min period; and

   b. 7 x 50 min periods – EC (total time).

6. Training and Learning Aids: Scenario (operational mission to be assigned).

7. Remarks: nil.

PO 003 – DEVELOP SONAR TEAM IN SUPPORT OF OPERATIONS

EO 003.01 DEVELOP SONAR TEAM IN SUPPORT OF OPERATIONS

1. Performance: Develop sonar team in support of operations.

2. Conditions:

   a. Given:

      (1) Training activity (refresher, familiarization, CRRs);

      (2) Training activity template;

      (3) DWAN;

      (4) CSNI; and

      (5) References and publications;

   b. Denied: Nil;

   c. Environment: Classroom.

3. Standard: The trainee shall develop sonar team in support of operations to include:

   a. Supporting Command with individual unit/ships packages to include:

      (1) Conducting Unit On the Job Training (OJT), Ship's Development Package (SDP) and Pre-Information Package (PIP) (G0085);

      (2) Compiling/distributing Unit OJT, SDP, PIP reports (G0040) to include, but not limited to:

         (a) Submarine safety IAW A10;

         (b) Basic Submarine Qualification; and

         (c) Submarine Sonar Conversions;

      (3) Compliance monitoring; and

      (4) Compliance enforcement;

   b. Planning/designing training exercises (G0010) to include:

      (1) Inputting training scenarios (G0090);

      (2) CRRs; and

MARPAC RDIMS #1387867
Chap 1 & 2 approved 01/12/2023
Chap 3 & 4 approved 18/01/2024

(3) Recommending improvements to training (L0035);

c. Conducting continuation/proficiency training (G0080);

d. Conducting team training events (G0065) to include:

(1) Assessing team training performance (G0120);

(2) Correcting team training deficiencies (G0070); and

(3) Providing section/unit with feedback (G0140);

e. Conducting individual training events (G0050) to include;

(1) Assessing individual training performance (G0115); and

(2) Correcting individual training deficiencies (G0055).

4. Teaching Points:

Teaching Points	Instructional Strategies	Time	References/Remarks
1. Identify and discuss UC1 roles and responsibilities. - Introductions - NFS(A) Sub Div Staff - MOC Advisor	Method: Guest Lecture Media: Instructor Environment: SECRET Classroom	3 x 50 min	
2. Contribute to intelligence gathering operations. - Trinity - METOC	Method: Field Trip Media: Guest Lecture Environment: Briefing room	2 x 50 min	
3. Identify services available to support submarine sonar operations. - CFMWC	Method: Guest Lecture Media: Guest Environment: SECRET Classroom	1 x 50 min	

Teaching Points	Instructional Strategies	Time	References/Remarks
4. Identify and discuss terms of reference for UC1s. - SSOs - CSOs - COTMs - CRRs/CFCD - DAODs - RCN Guide to the Divisional System	Method: Research Activity Media: DWAN Environment: Classroom	1 x 50 min	A10, A11, A15, A16, A17, A18
5. Develop sonar operators in support of operations.	Method: Case Study Media: Scenario Environment: Classroom	2 x 50 min	A10, A11

5. Total EO Time: Total: 11 x 50 minute periods as follows:

   a. 9 x 50 min period; and

   b. 2 x 50 min periods – EC (total time).

6. Training and Learning Aids:

   a. Training Activity Case Study; and

   b. Training Activity Template.

7. Remarks: Guest lecturers need to be booked in advance with Trinity, METOC, CFMWC, and NFS(A) Sub Div Commander.

QUALIFICATION STANDARD ANNEXES

MARPAC RDIMS #1387867
Chap 1 & 2 approved 01/12/2023
Chap 3 & 4 approved 18/01/2024

ANNEX A – MAIN REFERENCES

1. This annex lists all references used in this QS and TP.

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A		**Official CAF Publications and Directives that relate to this Qualification**
A01	NCPM 401 B	Victoria Class Combat Information Organization
A02	CFCD 106	Maritime Tactical Instructions
A03	NAVORD 2000-1	Intelligence Support to HMC Ships and Submarines
A04	CFCD 107	Communication Orders
A05	NAVORD 3395-1	Naval Signature Management Policy Organization
A06	NCPM 403	Submarine Sonar Operating Manual
A07	4A	Active Classification Guide
A08	CFCD 113D	Acoustic Information Guide
A09	MILOC Manual	Military Oceanography Manual
A10	VCSSO's	Victoria Class Ships Standing Orders
A11	CSO's	Captains Standing Orders
A12	NPTORD 4500-54	Training Review Board Process
A13	CFCD 109	Naval Signature Management Manual
A14	NCPM 414	Canadian Submarine Records Manual
A15	CFCD 102	Maritime Command Combat Readiness Requirements
A16	DAOD	Defence Administrative Orders and Directives
A17		Guide to the Divisional System
A18	COTM	Commanding Officer Temporary Memorandum
A19	VCEOP's	Victoria Class Emergency Operating Procedures
A20	CFCD 110A	Submarine Operating Instructions
A21	CFCD 119	Canadian Submarine Towed Array User Guide
B		**(Non-CAF) Government of Canada Publications & Foreign Military (Non-DND) Publications that relate to this Qualification**
B01	BR 2000(63)	British Reference
B02	AXP-1 E	Allied Submarine and Anti-Submarine Manual
C		**Civilian Publications and Manuals that relate to this Qualification**
C01	Nil.	
D		**Other Publications and Manuals that relate to this Qualification**
D01	Nil.	

NOTE: If additional references are identified during the TPWB this annex can be updated secretarially through NPTG HQ.

ANNEX B – TASK LIST

1. The task list includes all of the tasks associated with this qualification, and the outcome of the Train / No-train decisions. If a task is designated as requiring training, the PO where it is covered is indicated. If a task is designated as not requiring training, the justification for that designation is provided.

2. The following table defines some of the criteria used to support a no-train designation in the task list. Other reasons may be used if required.

CRITERIA	DEFINITION
A	**Previously qualified.** Tasks and knowledge already possessed by the student by CAF and/or RCN courses, training and/or experience.
B	**Will receive on future courses.** Tasks and knowledge will be covered as part of other CAF and/or RCN courses and/or training.
C	**Difficulty.** Task is not difficult to learn; formal training is not required.
D	**Importance.** Task is not important; it has no significant impact on mission success.
E	**Frequency.** Task is rarely performed, no formal training is required.
F	**Lack of equipment.** Currently there is no equipment in the CAF inventory to train this task.

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
A0005	Perform communications circuit functional checks (comm checks)	NT		Previously qualified BSQ. Simple.
A0010	Operate communications equipment	NT		Previously qualified BSQ.
A0020	Perform Search and Rescue (SAR)/Distress/SUBSMASH communications procedures	NT		Previously qualified BSQ.
A0025	Encode/Decode tactical information (with Authenticate communication transmissions)	NT		Previously qualified RQ S3.
A0030	Authenticate communication transmissions	NT		Previously qualified RQ S3.
A0035	Extract information from COMPLANS, EMCON Plans and Frequency Plans	T	001	Difficult and Important.

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
B0010	Plot formations, screens and dispositions	NT		Previously qualified AILO.
B0015	Maintain stateboards	NT		Previously qualified AKYG.
B0020	Enforce Emission control (EMCON) restrictions	NT		Previously qualified AKYG.
B0025	Adhere to Emission control (EMCON) restrictions	NT		Previously qualified RQ S3.
B0030	Contribute to intelligence gathering Ops	T	001	Importance, frequency, new context. High priority, proficiency level 4.
B0035	Display intelligence via UWW brief	T	001	Importance, frequency, new context. High priority, proficiency level 4.
B0040	Disseminate intelligence information	T	001	Importance, frequency, new context. High priority, proficiency level 4.
B0045	Identify Capacities and limitations of Platforms	T	001	Importance, frequency, new context. High priority, proficiency level 4.
B0055	Maintain operational logs	T	001	Importance, frequency, new context. High priority, proficiency level 4.
B0060	Assist with tests and trials with outside agencies	T	001	Importance, frequency, new context. High priority, proficiency level 4.
B0085	Conduct escape/rescue training	NT		Previously qualified BSQ.
C0005	Supervise HMS operations in a mine avoidance environment	NT		Nil requirement.
C0010	Supervise mine detection team in a mine avoidance environment	NT		Nil requirement.
C0025	Operate ancillary acoustic equipment	NT		Previously qualified AILO.
C0035	Launch underwater expendables	NT		Previously qualified AILO.
C0040	Analyzing XBT/XSV Forms	NT		Previously qualified AILO.
C0065	Analyze passive acoustic data	NT		Previously qualified RQ S3.
C0070	Track acoustic contacts	NT		Previously qualified RQ S3.
C0080	Classify passive acoustic data	T	002	New context. Low priority. Proficiency level 3.

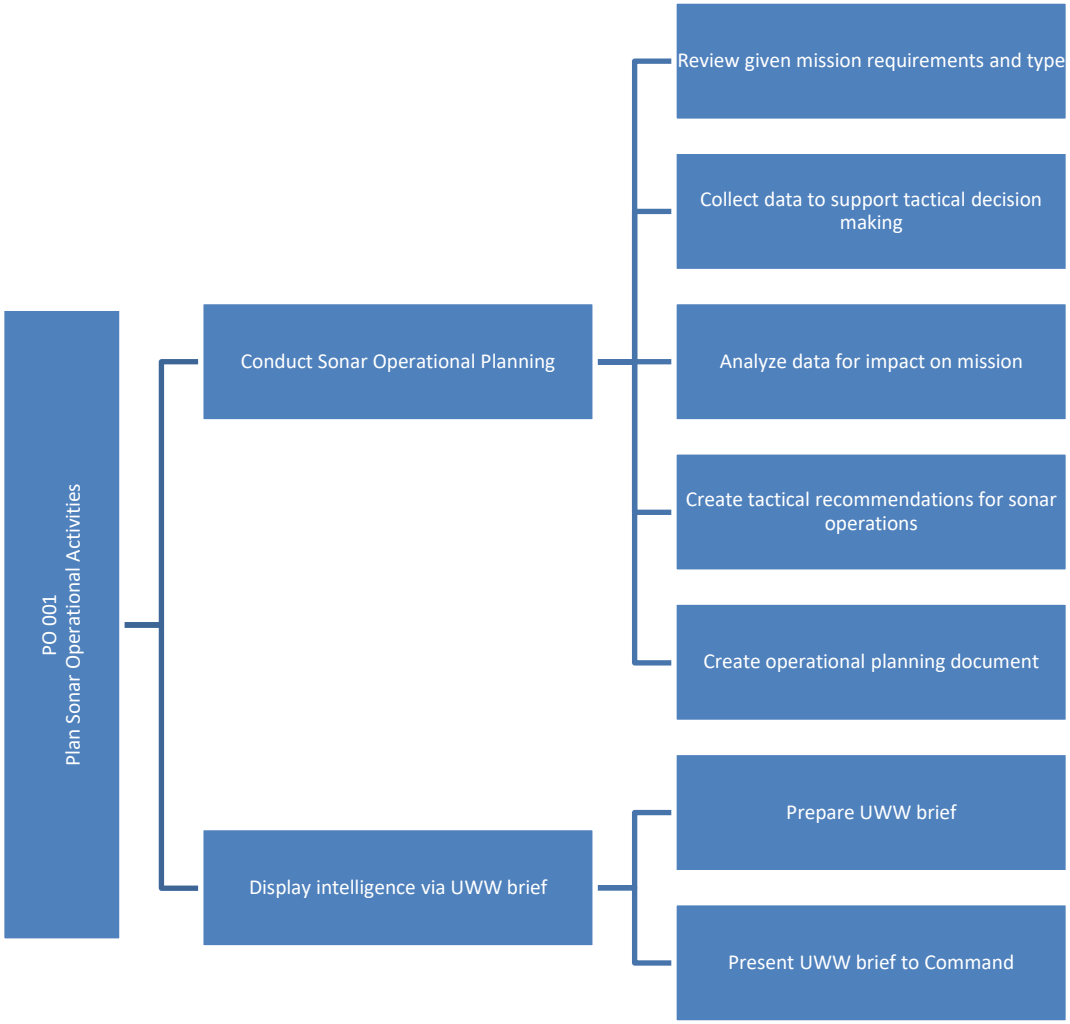
Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
C0085	Localize passive acoustic contacts	T	002	New context. Low priority. Proficiency level 3.
C0100	Construct Contact Area of Probability (AOP)	NT		Previously qualified AILO.
C0105	Correlate active/passive/non-acoustic information	T	002	New context. Low priority. Proficiency level 3.
C0115	Calculate acoustic sensor range information	T	002	New context. Low priority. Proficiency level 3.
C0120	Calculate array tow speeds, tilt and/or balancing	T	002	New context. Low priority. Proficiency level 3.
C0130	Conduct noise rounds	T	002	New context. Low priority. Proficiency level 3.
C0135	Perform noise rounds	NT		Supervise, not perform.
C0145	Coordinate ASW	NT		Should be "coordinate UWW".
C0150	Determine Oceanographic Effects on Acoustic Operations	T	001	New context. Moderate priority. Proficiency level 3.
C0155	Direct towed array system operations	T	002	New context. Moderate priority. Proficiency level 3.
C0165	Coordinate water space management	NT		Future qualification and mission workups.
C0170	Assist with loading/handling of torpedoes	NT		Previously qualified AINA.
C0195	Perform Passive Aural Listening (PAL)	NT		Previously qualified RQ S3.
C0200	Provide data for RFIs	T	001	Important, frequent. Moderate priority. Proficiency level 3.
C0205	Develop submarine special operations plans	T	001	Important, infrequent. Low priority. Proficiency level 1.
C0225	Operate Submarine Fire Control System (SFCS)	NT		Not a requirement.
C0235	Monitor operation of all submarine sonar equipment	T	002	Important, frequent. Moderate priority. Proficiency level 4.
C0240	Supervise Ship Control Console	NT		Previously qualified AILR.
C0245	Operate electronic fixing aids (EFAs)	NT		Previously qualified AILR.
C0265	Set up the Acoustic Positioning System (APS)	NT		Equipment not used.
C0280	Operate detection console	NT		Equipment not used.

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
C0285	Operate classification console	NT		Equipment not used.
D0045	Prepare UWW exercise support requests	T	001	Frequent, not difficult. Low priority. Proficiency level 2.
D0050	Assess UWW threat information	T	001	Important, Frequent, new context. High priority. Proficiency level 4.
D0055	Maintain quiet running bill	T	002	Important, Frequent, new context. High priority. Proficiency level 4.
D0060	Develop noise monitoring plan	T	001	Important. Moderate priority. Proficiency level 4.
D0065	Apply Water Space Management	T	001	Important, Frequent, new context. High priority. Proficiency level 4.
D0070	Optimize ship's acoustic signature	T	002	Important, Frequent, new context. High priority. Proficiency level 4.
D0125	Operate Submarine Fire Control System (SFCS)	NT		Not required.
E0005	Compile ASW tactical picture	NT		See ROD notes.
E0015	Detect radar contacts	NT		Not completed at this level.
E0020	Amplify track information	T	002	Important, Frequent, new context. High priority. Proficiency level 4.
E0030	Operate Submarine Fire Control System (SFCS)	NT		Not required.
F0005	Evaluate correlated acoustic information	T	002	Important, Frequent, new context. High priority. Proficiency level 4.
F0015	Identify changes in contact's readiness and tactical states	NT		Previously trained RQ S1.
F0045	Interpret oceanographic data	T	001	Important, Frequent, new context. High priority. Proficiency level 4.
G0010	Plan / design training exercises	NT		Post-board decision based on NPTG feedback on POs.
G0040	Compile / Distribute Unit On Job Training (OJT), Ships Development Package (SDP), Pre-Information Package (PIP) reports	T	002	Important, frequent, but with task delay tolerance. Low priority. Proficiency level 3.

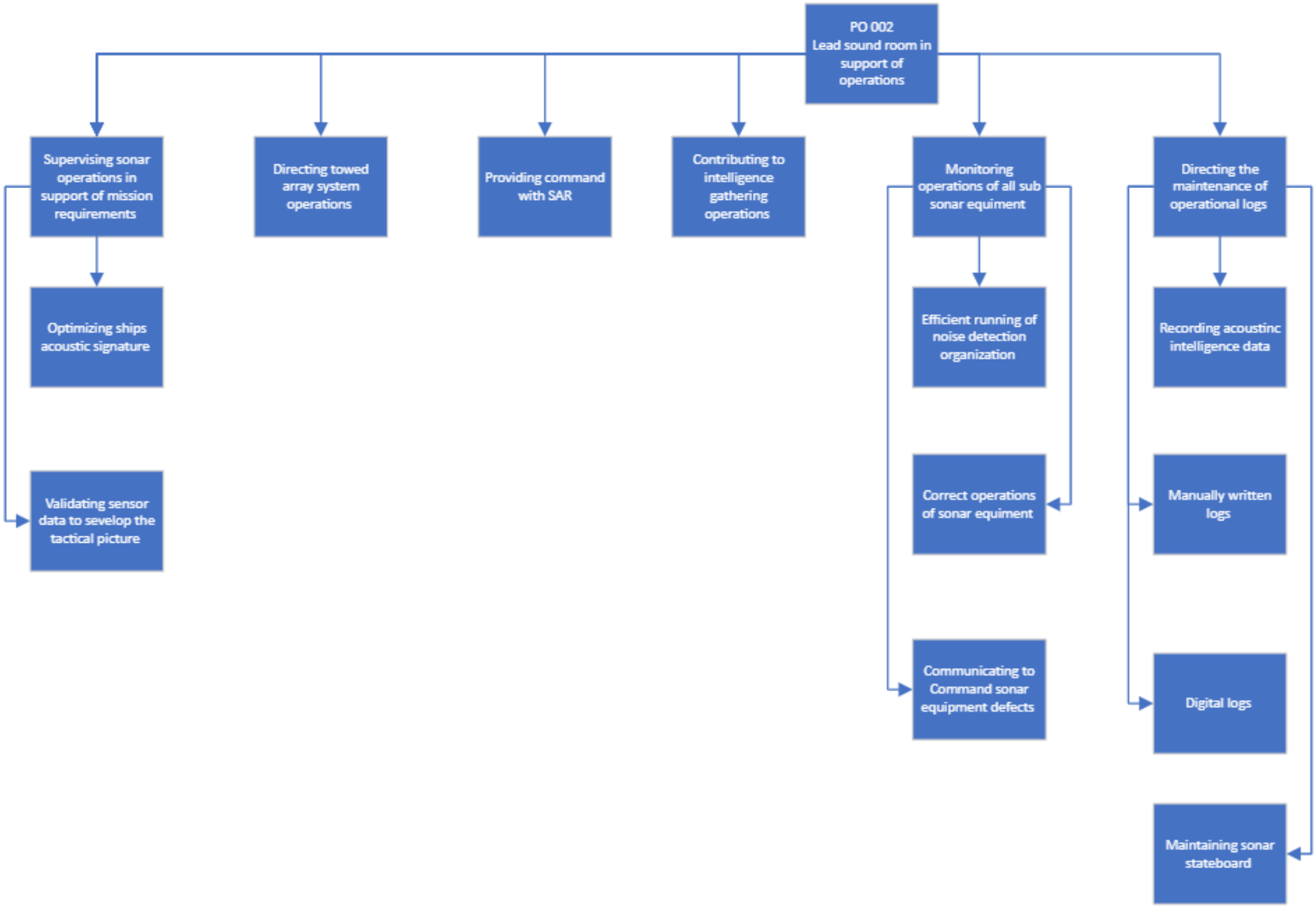
Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
G0045	Assist in individual training events	T	002	Important, frequent, but with task delay tolerance. Low priority. Proficiency level 3.
G0050	Conduct individual training events	T	002	Important, frequent, but with task delay tolerance. Low priority. Proficiency level 3.
G0055	Correct individual training deficiencies	T	002	Important, frequent, but with task delay tolerance. Low priority. Proficiency level 3.
G0060	Assist in team training events	T	002	Important, frequent, but with task delay tolerance. Low priority. Proficiency level 3.
G0065	Conduct team training events	T	002	Important, frequent, but with task delay tolerance. Low priority. Proficiency level 3.
G0070	Correct team training deficiencies	T	002	Important, frequent, but with task delay tolerance. Low priority. Proficiency level 3.
G0080	Conduct continuation / proficiency training	T	002	Important, frequent, but with task delay tolerance. Low priority. Proficiency level 3.
G0085	Conduct Unit On Job Training (OJT), Ships Development Package (SDP), Pre-Information Package (PIP)	T	002	Important, frequent, but with task delay tolerance. Low priority. Proficiency level 3.
G0090	Input training scenarios	T	002	Important, frequent, but with task delay tolerance. Low priority. Proficiency level 3.
G0095	Operate simulator/trainer systems	NT		Not completed as UC1, is tasking for different context.
G0100	Provide recommendation to Progress Review Boards (PRBs) / Training Review Boards (TRBs)	NT		NPTG recommended this task NT.
G0115	Assess individual training performance	T	002	Important, frequent, but with task delay tolerance. Low priority. Proficiency level 3.
G0120	Assess team training performance	T	002	Important, frequent, but with task delay tolerance. Low priority. Proficiency level 3.
G0140	Provide section/unit with feedback	T	002	Important, frequent, but with task delay tolerance. Low priority. Proficiency level 3.

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments
L00005	Conduct Programme evaluation (alignment of docs, availability of trg materials	NT		Instructor role.
L0010	Administer course critique process	NT		Instructor role.
L0015	Track and monitor action items identified during course evaluation	NT		Instructor role.
L00035	Recommend improvements to training	T	002	Important, frequent, but with task delay tolerance. Low priority. Proficiency level 3.

ANNEX C – QS SCALAR



MARPAC RDIMS #1387867
Chap 1 & 2 approved 01/12/2023
Chap 3 & 4 approved 18/01/2024



MARPAC RDIMS #1387867
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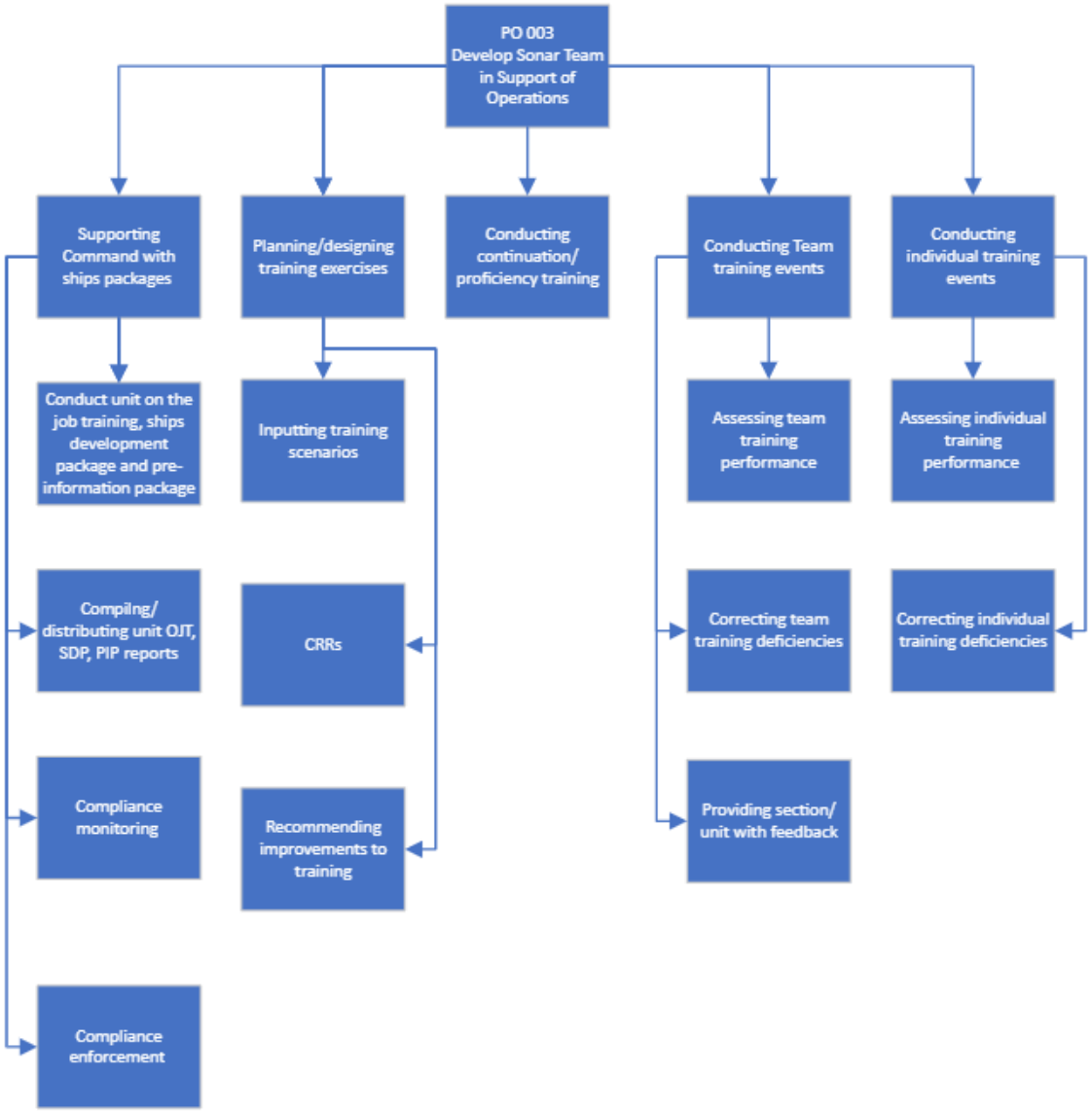

MARPAC RDIMS #1387867
Chap 1 & 2 approved 01/12/2023
Chap 3 & 4 approved 18/01/2024

ANNEX D – RECORD OF DECISIONS QSWB

1. This record summarizes the conduct of the QSWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

Title	**Submarine Sonar Underwater Controller (Underwater Controller One) QSP WB**			
Location of Board	**CFB Halifax**			
Dates	**21 Feb – 4 Mar 22**			
Board Composition				
	Surname	Initials	Rank	Unit
Co-Chair (content)	Larkin	MD	PO1	NTDC(A)
Co-Chair (process)	Hart	TD	Lt(N)	NTDC(A)
TM	Shewchuk	SJAM	PO2	NFS(A)
SME	Holland	STM	PO2	HMCS WINDSOR
SME	Rogers	JBM	MS	HMCS WINDSOR
SME	Leveille	SM	MS	NFS(A)
SME	Larkin	MD	PO1	NTDC(A)

Day #	Details	Action
1	The purpose of convening this working group was to determine the on-job performance requirements and training and assessment design for Submarine Sonar Operator Underwater Controller 1. Specification Task – No Train Decisions: See Annex B.	
2	Upon review of existing and current Unique Specialty Qualification specification, the Occupational Specification, and the current QSP, it was determined that there is a disconnect between these documents and the career progression of Submarine Sonar Operators. It was discussed with NTDC(A) Sub Div and other stakeholders to confirm that a MESCR was drafted for submission to have the AILE (UC1) course be the equivalent to the Sonar Op PO2 RQ for Submarine Sonar Operator, as per trade progression changes enacted in 2012 precluding Submarine Sonar Ops from the requirement for the surface courses RQ-PO2 and RQ-PO1. Further actions post-board to have this MESCR staffed and approved will be required, however the board is creating this QSP to reflect all tasks required for the PO2 Sub Sonar operator as if it	

Day #	Details	Action
	were an RQ and not simply a USQ additional to an RQ.	
3	The board conducted a Train / No-Train task disposition based on the Occupational Specification Master Task List (MTL) for the Subs SCS position, which is the basis for the PO2 RQ for Submarines. This task disposition was completed using the criteria from the following performance supports, as per embedded file in Annex B.	
4	Through task analysis, the board has decided that there will be two Performance Objectives (POs), Plan Sonar Operational Activities, and Lead Sonar Operators in Support of Operations.	
5	The board reviewed multiple sources of criteria relevant to oral communication, briefings, and recommendations to come up with detailed criteria required for on the job performance of UC1s for planning and briefing recommendations to Command. These criteria are detailed at a higher level than previous qualifications as the UC1 is expected to provide detailed, justifiable, and in-depth recommendations at a high level.	
6	ASW vs. UWW tactical picture, the board deliberated on task E0005 “compile ASW tactical picture” at length. It was noted that Submarine Sonar operations are not limited to ASW, and that the task is not inclusive of all duties required for UC1 or sonar operators. The board has decided to change this to a “no train” task and will re-word the required performance in PO 002 as “oversee the compilation of the tactical picture” as this better reflects the requirements of the UC1 and broader tactical picture requirements. It is noted that the compilation of the picture is the responsibility of the Sonar Watch Leader, however the UC1 is responsible to confirm the picture, analyze the data, and assess the situation based on the tactical picture.	
7	Conditions: the board spent significant time reviewing the general context and conditions of the performance objectives. In addition to the conditions listed in the performance objectives, the board will add additional guidance.	
8	Working conditions from the AILE specification: “The working environment encountered includes all conditions applicable to the occupation. In addition,	

Day #	Details	Action
	employment in submarines provides a set of working conditions for which there is no equivalent in the surface fleet. Submarine duties involve an extremely restrictive environment where noise, extremes of temperature, high humidity, atmospheric contaminants, fluctuating atmospheric pressure and lack of sleep contribute to high levels of physical strain and fatigue. The requirement to live and work for extended periods in close quarters with a large number of other personnel in cramped, unpleasant conditions creates a high degree of emotional stress, which is exacerbated by the fact that a simple error at any time while the submarine is submerged could have catastrophic consequences for the entire crew. Submarine duties may lead to physical disorders arising from extended periods of exposure to atmospheric contaminants such as high partial pressures of carbon dioxide and significant concentrations of carbon monoxide, hydrocarbons, chlorine, stibine, arsine and the combustion products of freon. Reduction in sight efficiency may result from extended periods of exposure to inadequate illumination and reduction in hearing efficiency may result from exposure to engine and ventilation system noise and rapid changes in air pressure.”	
9	Additional conditions guidance for each PO is outlined in Tables 1 and 2 below. Note: these are recommended for use in justifying qualitative or subject assessment criteria during the training course and also for the assessment of learning disabilities.	
10	Assessment criteria for PC 002 and ECs 002.02 and 002.03: the board purposely left the assessment criteria on these checklists as very broad. The UC1 is required to act at a high standard in many situations with little time to gather the required information and make recommendations. As such, these assessments list the factors that are required for the UC1 to take into account when making recommendations to Command while at sea, but cannot list the outcomes of these recommendations nor the exact steps that are taken in order to provide these recommendations. As these assessments require the UC1 to react to multiple situations during the assessment period, the decision of “SAT” or “UNSAT” should be based on the overall performance of the UC1 during all of these evolutions.	

Day #	Details	Action
	As additional guidance, the board suggests that the criteria for recommendations to command include: timely recommendations based on immediately available input. Further clarification can be sought and recommendations amplified or updated as required based on additional information received.	
11	Definition of “tactical posture”: in PO 002, the sub-task of “making recommendations for movements to optimize tactical posture” is meant to cover submarine movements and speeds that will affect the acoustic signature of the submarine. The UC1 is required to take into consideration the operations of an SSK based on the capacities of limitations of the platform. It is expected that the UC1 will recommend movements and speeds in consequence to required “tactical posture”, defined by the board as “priority and tactics for achieving mission based on considerations for detection, counter-detection, and area of operations”.	
12	Note on “tactics” and emphasis of decision making for UC1s: based on SME input and recent graduates of the UC1 course, the board discussed that there is a “desire” within the Submarine Sonar community to gain more in-depth knowledge and practice on tactical decision recommendations during the UC1 course. Given UC1s are required to make recommendations on the job to Command concerning movement, operations, and tactics for the submarine, the board has focused on discussion of tactics and implication of available data within EO 001.01 through in-depth discussions and feedback. Additionally, there are several caveats within this QSP that specify that decisions must be made based on available information, and can be amplified and clarified as the situation develops. While there is no available doctrine specifically for Sub Sonar Ops, the board has endeavoured to ensure that there is ample opportunity to discuss and practice the application of various tactics throughout this QSP.	
13	Trainee CSNI accounts: the board notes that for trainees coming on this course, they will require a CSNI account. NFS should ensure that as much advanced notice (through joining instructions and pre-information packages) to trainees is provided to have an account created.	

Day #	Details	Action
14	Mine warfare, mine avoidance/detection: the board discussed at length whether mine warfare, mine avoidance and mine detection were required portions of the UC1 job and this QSP. The board has considered this and decided that at this time, there is not supporting doctrine, operational capacities, nor available expertise to include mine warfare related to Victoria Class Submarines within this QSP. It is advised that there will be the opportunity to have discussions and briefings on mine warfare from subject matter experts from RCN on the job, that will assist in covering this gap.	
15	Template: “operational planning document” the board is suggesting a template for an operational planning document in PO 001. It is noted that there is no standardized template for briefing or planning that is used across the Sub Sonar community, however the board is providing this template as a training aid to give trainees an opportunity to organize and track data as well as demonstrate recommendations. This document is not required for the overall performance or assessment in this QSP, however it is absolutely expected that the trainee will be able to organize and provide amplification on data used for recommendations and the board suggests that this template will assist trainees in doing this. If trainees choose to maintain their own filing system for this data and their resultant recommendations, it should be considered as acceptable.	
16	Additional divisional responsibilities required at the UC1 level: the board has identified that there are multiple divisional responsibilities for the UC1 that will not be covered in this QSP, but may be available through other training: PER/PDRs (through PACE training), staff work (PLQ), and disciplinary and administrative actions which may be touched on in PLQ but have other available sources of performance support, such as military justice at the Summary Trial manual.	
17	PC 001: the board has chosen to do a BARNSTORM situation for this assessment, as it reflects the requirement of the UC1 to perform under intense pressure on the job and provide situation, assessment, and recommendations to Command based on the skills and knowledge gained during PO 001. It is	

Day #	Details	Action
	expected that this is more reflective of on-job requirements than a long-mission planning scenario.	
Post-Board Addendum		
18	Of note on course capacity: the board chose to maintain the previous course capacities for minimum, optimum, and maximum load. It is not expected that maximum load will ever be achieved, but optimum and minimum loads were confirmed as logical based on course content and available facilities (e.g. TWS trainer).	
19	Pre-requisites: it is noted that the JBS specification pre-requisites (be SS AILS qualified, be SS AILO qualified, be QL6A qualified, be the rank of MS) do not match the pre-requisites that are listed in this QSP. A MESCR has been staffed to have the specification itself altered, given that it does not match the career path of Submarine Sonar Operators, and the qualifications listed are outdated.	
20	Follow up actions: See Table 3 below.	

Table 1 : Conditions Guidance – PO 001

Conditions		
Category	Factors	Notes
Distractions	Noise	Office environment, nil overload expected
	Smell	Office environment, nil overload expected
	Lighting	Office environment, nil overload expected
	Space issues / clutter	Office environment, nil overload expected
	Visual distractions	Office environment, nil overload expected
	Sensory load (overload)	Office environment, nil overload expected
	Multiple things happening at once (e.g. movement, sounds, lights, distractions)	Group office setting, it is expected that there will be distractions, requests and little ability to focus on only one task without interruption
	Seating / standing arrangements	As preferred
	Group vs. Solo activity	Production of plan is responsible to UC1, however the work will need to be done in groups, delegated, and synthesized
Time	Scheduling (schedule available)	Schedule is expected to be available, but is expected to be

		in flux with other activities injected
	Breaks available / continuous work required	Breaks available
	Processing time / immediate response	Processing time is available
	Transition time between activities	Available
Physical	Standing / seated in one position (freedom of movement)	Movement is relatively free
Interaction	Assistance available (peer / buddy system / supervisor)	Peer and subordinate assistance avail, able to ask network and CoC for clarifications and input
	Supervision available	Yes, but is expected to be entirely self-directed
	Feedback available (immediate or delayed)	Immediate feedback is generally avail
	Facilitation available	N/A
	Cooperative vs. competitive environment	Cooperative
	Are job aids / performance supports / references available	References are available, templates differ, investigation and self-directed analysis required for performance
Cues / Triggers	Visual cues (signal, direction/signage)	N/A
	Auditory / oral (alarms)	N/A
	Direct vs. indirect tasking (clear, concise, unambiguous)	Direct tasking expected, experience is expected to inform additional actions required to supplement direction
	Social cues	N/A
	Other cues	N/A

Task Analysis & Standards		
Category	Factors	Notes
Media	Written (typed/handwritten)	Both
	Auditory / oral	Some portions of direction will only be orally communicated
	Visual (graphics/tables, multimedia)	A lot of information provided/gained will be visual
	Recordings available	No
	Scribing available / minutes / synopsis	No
	Scenarios available	Yes
	Adaptive media available (e.g. speech-to-text)	No
	Are examples (of completed work) available	Yes
Time / Task management	Structured vs. unstructured task	Loosely structured, with some baseline standards
	Alarms / reminders available	Self-directed only
	How is task received	Orally and/or written
	Is time allotted to process / read / review information	Yes
	Is time allotted to understand / reflect on information	Yes
	Is time allotted to research	Yes
	Is time allotted for confirmation / clarification	Yes
	Are there time constraints for the performance of the task (to do the task)	Timelines and deadlines are set in tasking
	Is advanced notice / warning issued for the task	Yes
	Is continuous performance a requirement (no breaks)	No
	Is an immediate reactionary response required	No

	Is time allotted for practice	Yes
	Is trial / error and on-job learning expected / allowed	Expected, experience will increase performance, but always expected to be learning
	Is the intended outcome / goal evident	Yes
	Is the consequence for error evident / clarified	Yes
	Is there the ability to “opt out” of all, or portions of, the task	Delegation and assistance are available, however UC1 is responsible for the output
Prioritization	Is the priority self-evident/clear or ambiguous	Sometimes clear, self-prioritization of efforts is required, and analysis of what will be required to meet mission is required of UC1
	Is priority imposed or self-directed	Often prioritization is self-directed based on timelines, effort required, and expected mission requirements
Interaction	Are clarifications (from supervisor, tasker, references, or peers) available	Yes
	Is there a level of social competence involved / expected	Yes, high-level required for interactions with Command and for supervision of subordinates and liaison with out-stations
	Is positive reinforcement or punishment used to reinforce expected behaviours	Immediate reaction to product may be harsh
	Is feedback available (immediate vs. delayed, source, and level of clarity)	On final product: immediate During product development: delayed and peer-sourced generally
	Are job-shadowing / task rehearsal available	Yes
	Is troubleshooting assistance available, or only self-reliance	Assistance available
	Is progress monitoring / oversight expected	Majority of task is self-directed without direct oversight
Task Complexity	Is the task clear vs. ambiguous (is judgement required and is nuance present)	Product requirement is clear, pieces to be included are ambiguous and require judgement
	Are there numerous possible outcomes	Yes
	Are the steps in the task clear, concise, simple, and easy to follow	Loosely laid-out process, uses judgement and experience to create own SOPs for conduct
	Can the task be broken down into or completed in “chunks”	Yes
	Does the task involve changing parameters (frequency and severity of change)	Multiple changes are expected, changes may be fundamental and require re-start of task
	Are expectations clear	Often vague
	Are directions clear	Often vague

Table 2 : Conditions Guidance – PO 002

Conditions		
Category	Factors	Notes
Distractions	Noise	In-line with specification working conditions
	Smell	In-line with specification working conditions
	Lighting	In-line with specification working conditions
	Space issues / clutter	In-line with specification working conditions
	Visual distractions	In-line with specification working conditions
	Sensory load (overload)	In-line with specification working conditions
	Multiple things happening at once (e.g. movement, sounds, lights, distractions)	Many distractions are expected to be happening at all times.
	Seating / standing arrangements	In-line with specification working conditions
	Group vs. Solo activity	Will always be working in a group environment, in close quarters, even when assigned a solo activity.
Time	Scheduling (schedule available)	Schedule requires extreme flexibility, can be required to immediately change activities with no notice
	Breaks available / continuous work required	There are occasions where no breaks will be available for extended periods
	Processing time / immediate response	Immediate responses are frequently required
	Transition time between activities	There may be transition time between activities, but it is normally not available
Physical	Standing / seated in one position (freedom of movement)	Freedom of movement is limited, given small quarters, and no ability to leave the submarine
Interaction	Assistance available (peer / buddy system / supervisor)	Yes, though UC1 is ultimately responsible for recommendations made
	Supervision available	No
	Feedback available (immediate or delayed)	Immediate feedback is available
	Facilitation available	No
	Cooperative vs. competitive environment	Cooperative
	Are job aids / performance supports / references available	Limited availability, no connectivity to outside
Cues / Triggers	Visual cues (signal, direction/signage)	Many competing visuals
	Auditory / oral (alarms)	Many competing auditory signals
	Direct vs. indirect tasking (clear, concise, unambiguous)	Generally direct tasking
	Social cues	N/A
	Other cues	N/A

Task Analysis & Standards		
Category	Factors	Notes
Media	Written (typed/handwritten)	Both
	Auditory / oral	A lot of oral communication and direction
	Visual (graphics/tables, multimedia)	Highly visual
	Recordings available	Yes
	Scribing available / minutes / synopsis	No
	Scenarios available	Yes
	Adaptive media available (e.g. speech-to-text)	No
	Are examples (of completed work) available	Yes
Time / Task management	Structured vs. unstructured task	Structured at sea
	Alarms / reminders available	Yes, for overall activities of Submarine
	How is task received	Oral and written direction
	Is time allotted to process / read / review information	Sometimes, but often required to process immediately
	Is time allotted to understand / reflect on information	Sometimes, but often required to process immediately
	Is time allotted to research	Sometimes, but often required to process immediately
	Is time allotted for confirmation / clarification	Sometimes, but often required to process immediately
	Are there time constraints for the performance of the task (to do the task)	Yes, always
	Is advanced notice / warning issued for the task	Sometimes, however immediate response to developing situations or incoming requests from Command are frequently required
	Is continuous performance a requirement (no breaks)	Sometimes
	Is an immediate reactionary response required	Frequently
	Is time allotted for practice	Yes, scheduled practice of some performance requirements is available
	Is trial / error and on-job learning expected / allowed	At sea, responsibility and proficiency requirements are extremely high and have major consequences
	Is the intended outcome / goal evident	Yes
	Is the consequence for error evident / clarified	Yes
	Is there the ability to “opt out” of all, or portions of, the task	No
Prioritization	Is the priority self-evident/clear or ambiguous	Clear
	Is priority imposed or self-directed	Imposed
Interaction	Are clarifications (from supervisor, tasker, references, or peers) available	Yes
	Is there a level of social competence involved / expected	High level of interactions during long periods of emotional and physical strain
	Is positive reinforcement or punishment used to reinforce expected behaviours	Both are used, depending on situation
	Is feedback available (immediate vs. delayed, source, and level of clarity)	Immediate feedback is generally available

	Are job-shadowing / task rehearsal available	Sometimes at sea this is available, but only rarely
	Is troubleshooting assistance available, or only self-reliance	Yes
	Is progress monitoring / oversight expected	Yes at sea
Task Complexity	Is the task clear vs. ambiguous (is judgement required and is nuance present)	Clear taskings
	Are there numerous possible outcomes	Yes
	Are the steps in the task clear, concise, simple, and easy to follow	Yes
	Can the task be broken down into or completed in “chunks”	Yes
	Does the task involve changing parameters (frequency and severity of change)	Frequent and drastic changes expected with short/no notice
	Are expectations clear	Yes
	Are directions clear	Yes

Table 3 : Follow Up Action

Item	Action	OPI	Expected Date of Completion
1	Monitor MESCR submitted 27 Oct 2020 MARPAC RDIMS 1098556	NTDC(A) Sub Div Chief	Dec 2022
2	Follow-on courseware development	NTDC(A) CCD(A)	Apr 2023
3	Submit LSRs as per Annex G	NTDC(A) CCD(A)	Aug 2022

RECORD OF DECISIONS WORKING GROUP

Date: August 17-18 2023
Members:

- TDO Lt(N) Horvat
- SME CPO2 Laurin
- SME CPO2 Simoneau
- SME PO2 Holland

ITEM	DISCUSSION	ACTION
QS condition	TDO explained that much of the QS is built for training rather than on the job performance. Lt(N) Hart replied with feedback to questions prior to the working group being sat. Group met and was on board with adjusting the conditions and standard however the T/NT decisions were not going to be altered from the time the initial board was sat.	Reviewed comments, adjusted conditions (given, denied and environment)
Task: Identifying capacities and limitations of platforms (B0045)	The way that this task is written may need to be altered for future. The terms caps/lims resonate with the sonar operators, however the task itself it not reflective of the job. Nil action was taken at this time, but to make note for future.	Nil: Consideration for future
Access to request for information (RFI)	Although placed in the given category, the working group recognizes that it may not be a capability at this time, however will be available as updates are being made.	Added to Conditions: ‘Given’

Sonar equipment; Ancillary equipment;	This shall include major and minor equipment as well as any job aids that are used. As references and items get updated, this will ensure that all equipment will be covered.	Added to Conditions: 'Given'.
Tactical picture	TDO questioned what this means, however resonates in the sonar operating world as valid and representative.	Nil: Remains in as is.
Situation, Assessment, Recommendation (SAR)	This was identified as a standard way that briefing occurs WRT commands getting informed. This is be formalized by sea training via order. SME team reached out to sea training CPO2 Kistantiansen for details. Response received.	A02 represents the reference.
PO 003	Decision to add in another PO for developing the sonar team was made due to the importance and tasks that are included within the trade. Significant rework was completed to refine the tasks that were listed. Concerns were that the majority of these tasks were covered on their leadership training, however it was deemed that these tasks and this performance was important to have it be a stand-alone performance. Train/No Train decisions were not altered from initial board.	Split PO 002 into 2 separate. Leading sound room and developing the sonar team.
Para 6. REMARKS PO 003	TDO recommends that due to the nature of the performance, a summative assessment will not be possible on all items listed. Formative assessments will occur throughout the training of this performance.	TP board will development assessments as required.
Contributing to intelligence gathering operations (B0030)	TDO questioned the meaning of this task and SME team communicated that this task is very different than tasks listed in PO 001.	Remains in PO 002 as its own task.
Tasks	Reworked tasks were reviewed and updated to reflect its relevant PO. Task names were not altered.	Nil.
General modification	Reworked the QS to reflect what is done on the job versus training. Refined to the standard was not class specific.	Nil.
ROD	Reviewed by team, submitted at end of day 18 Aug.	Nil.

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the QS is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted QS Change Request (QSCR), thus ensuring the QSP has the most relevant information.

Version:	**1.01**
Date: 1 Dec 2023	**Who Initiated Change: NTG HQ**
Requirement for the Amendment: **1 PO deemed too large**	
Description of Changes: **1 PO to 3 PO's**	
Change Tracking number: **Nil**	

Trg Mgr: **Ops TM / NTG HQ / CPO2 Laurin**

TRAINING PLAN ANNEXES

MARPAC RDIMS #1387867
Chap 1 & 2 approved 01/12/2023
Chap 3 & 4 approved 18/01/2024

ANNEX E – RESOURCES, EQUIPMENT, SUPPLIES, AND ANCILLARIES

1. The following resources, equipment, supplies, and other elements are required in support of training. Resources are described in terms of formed organizations and are to arrive equipped with all personnel, vehicles, weapons, communications, and ancillary equipment required for conduct and integral support of operations described in the POs.

2. Facilities/Trainers

SER	DESCRIPTION		QTY	REMARKS
a.	VCSCTT / TWS		1	PO 002
b.	SECRET classroom		1	ALL POs
NOTES:				

3. Ammunition (Nil)

SER	LOG GUIDE	DESCRIPTION		QTY	REMARKS
a.					
b.					
c.					
NOTES:					

4. Weapons (Nil)

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

5. Specialized/Technical Equipment (Nil)

SER	NSN	DESCRIPTION	QTY	REMARKS

a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

6. Transport/Vehicles (Nil)

SER		NSN	DESCRIPTION		QTY	REMARKS
a.		XXX-XX-XXX-XXXX				
b.		XXX-XX-XXX-XXXX				
c.		XXX-XX-XXX-XXXX				
	NOTES:					

7. Communication Equipment/Frequencies (Nil)

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

8. Training Aids (Nil)

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

9. POL Products (Nil)

SER	NSN	DESCRIPTION	QTY	REMARKS

a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

10. General Purpose Stores (Nil)

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

11. Stationery (Nil)

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

12. Personal Equipment (Nil)

All students will have in their possession the personal equipment detailed in the course joining instructions. All personal equipment to be assigned to the students are accounted for in the resource tables above.

ANNEX F – COSTING INFORMATION

Estimates only, based on CF Cost Factors Manual

cost-factors-manual (mil.ca)

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: EXAMPLE		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	6	$1,500.0	[redacted]	$9,000	$9,000	$45,000
R&Q	6	$72.31	21	$9,111.06	$9,111.06	$45,555.30
Incidentals	6	$17.30	23	$2,387.40	$2,2387.40	$11,937.00
Subtotal				**$20,498.46**	**$20,498.46**	**$102,492.30**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	6	$50.00	1	$300.00	$300.00	$1,500.00
Instructional Materials (Printed / Handouts etc.)	6	$30.00	1	$180.00	$180.00	$900.00
Subtotal				**$480.00**	**$480.00**	**$2,400.00**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
	Total Estimated Costs			**$20,978.46**	**$20,978.46**	**$104,892.30**

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

1. This annex is to be used to properly schedule all required classrooms, trainers, ships, personnel, etc. for the duration of the course. Admin and PT periods are accounted for IAW NPTORD 4500-22 Annex B (QSP Instructions).

2. The tables are completed in the preferred sequence for lessons and resources. Special instructions to schedulers shall be included as "Notes" following the applicable PO or Module.

3. The table below is available to use as a template for calculations.

Allocation of ATT	**Minutes**
Actual Training Time	6,800 (136 pd)
ATT Sub-total	**6,800**
Allocation of BAT, TAT, TTT, PT (Course > 31 days)	
Base Administrative Time (BAT)	100
Training Administrative Time (TAT)	100
Training Travel Time (TTT)	600
Physical Fitness (PT)	600
Sub-total	**1,400**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **Total mins/400 = # days** (rounded up)	
Total time (minutes)	**8,200**
Total Training Days	**21**

ANNEX H – ASSESSMENT PLAN

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
EC 001.01	EO 001.01	To assess trainees ability to produce a sonar operational plan	Trainees will be assigned an operational activity and will need to create an operational plan for the sound room to meet the objective	Upon completion of EO 001.01	700 mins (14 pd) total - 50 min introduction 500 min planning -150 min group debrief	- Sonar operational planning template - Mission scenario	Nil	10/12 SATs	Remedial Training / Re-test
EC 001.02	EO 001.02	To assess trainees' ability to brief command on sonar operational plan developed in EC 001.01	Trainees will take mission plan created in EC 001.01, will create an UWW Brief, and will provide this brief to Command to	Upon completion of EO 001.02	350 mins (7 pd) total	UWW brief template	Nil	5/5 SATs	Remedial Training / Re-test

MARPAC RDIMS #1387867
Chap 1 & 2 approved 01/12/2023
Chap 3 & 4 approved 18/012024

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
			seek approval for their plan						
PC 001	PO 001	To assess trainees' ability to produce an operational plan under pressure in an operational setting	Trainees will be provided with a new mission requirement (BARNSTORM), and will be required to analyze current situation, gather information, create tactical recommendations, and brief command for approval of recommendations	Will be injected into EO 002.01 or 002.02 trainer activities.	150 min total (not additional time, to happen during assigned trainer time in PO 002	BARNSTORM scenario inject	Nil	14/16 SATs	IAW NTORD 4500-54
EC 002.01	EO 002.01	To assess trainees' ability to lead the sound room in support of operations	Trainees will act as UC1 in sound room and respond to operational taskings while leading the sound room team	Upon completion of EO 002.01	350 min (7 pd)	VCSCTT / TWS trainers	Nil	5/5 SATs	Remedial Training / Re-test
EC 002.02	EO 002.02	To assess trainees' ability to lead the sound room in support of	Trainees will act as UC1 in sound room and respond to operational taskings using towed array while	Upon completion of EO 002.02	350 min (7 pd)	VCSCTT / TWS trainers	Nil	6/6 SATs	Remedial Training / Re-test

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
		towed array operations	leading the sound room team						
PC 002	PO 002	To assess trainees' ability to lead the sound room as UC1 in support of operations	Trainees will act as UC1 in an operational sound room (trainer) and lead the sound room team	Upon completion of PO 002 / end of course	650 min (13 pd)	VCSCTT / TWS trainer	Nil	6/6 SATs	IAW NTORD 4500-54
EC 003.01	EO 003.01	To assess trainees' ability to develop sound room team to support operations	Trainees will be given a mission scenario and will have to develop a training activity to prepare / develop their sound room team to support this operation	Upon completion of EO 003.01 / Prior to commencement of EO 001.01	100 mini (2 pd)	Classroom, Training activity template	Trainees will not be required to run the training session or create the scenario inputs	5/6 SATs	Remedial training / re-test
PC 003	Nil	Nil	Nil	Nil	Nil	Nil	Nil	Nil	Nil

ANNEX I – ASSESSMENT CHECKLISTS

ENABLING CHECK 001.01 PREPARE TACTICAL RECOMMENDATIONS

EC Time: 550 mins (concurrent)150 min debrief

ASSESSMENT ACTIVITY: You are the UC1 for a Victoria Class Submarine. Command has provided a mission that will need to be achieved and it is your responsibility to plan and make recommendations for sonar operational activities that will achieve this mission.

This assessment will take place over 2 training days. Each trainees will be provided with a specific mission from “Command” (played by instructor or Combat Officer) and directed to produce a plan and recommendations for sonar operational activities that will meet the mission requirements. They will be provided with a template operational planning document and be allowed to ask clarification questions. (50 minutes)

Trainees will be required to asynchronously review mission requirements and establish a data collection plan. The trainees will be required to collect available open-source data and list required additional data. Additional data will have to be requested through the RFI process by drafting a, for training purpose, RFI message for submission to Trinity and the instructor. The instructor shall provide additional details from scenario as required, dependant on return from Trinity.

The instructor will provide requested data (or tell the trainee the particular data requested is unavailable) to support the scenario NLT end of Day 1.

Trainee will collate data and analyze it, make recommendations, and submit the operational planning document to the instructor NLT lunch on Day 2.

Instructor will review submitted plan, data collected, and recommendations, and provide feedback to each individual (afternoon, Day 2).

(500 minutes, 11 periods)

Individual feedback will be provided through the EC checklist in an individual meeting (afternoon of Day 2), where the instructor will review all components of the submitted operational planning document and provide input on recommendations. (150 minutes, 3 periods)

Pass/fail criteria: Trainees must submit an operational planning document with recommendations and data collected and must achieve 10/12 SATs on the EC checklist to pass this assessment.

ASSESSMENT RESOURCES:

- Operational Planning Document template
- Operational scenario

IMPACT OF FAILURE: Remedial Training / Re-test

INVIGILATOR NOTES:

- Trainees may collaborate on production of operational planning documents, however each trainee must submit an individual document. Trainees are additionally capable of using outside resources if they chose (e.g. “call a friend”).
- Limit instructor intervention during asynchronous activities, trainees should be self-sufficient for data collection and RFI.

1. PASS / FAIL CRITERIA: Trainees must submit an operational planning document with recommendations and data collected and must achieve 10/12 SATs on the EC checklist to pass this assessment.

2. IMPACT OF FAILURE: Remedial training, re-test.

Underwater Controller 1 (AILE) – EC 001.01 Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

ITEM	**PASS**	**FAIL**	**COMMENTS**
Plan sonar operational activities			
Review given mission requirements and type	☐	☐	
Collect data to support tactical decision making	☐	☐	
Identify and request additional data	☐	☐	
Analyze data for impact on operation			
Assess UWW threat information	☐	☐	
Identify capacities and limitations of platforms	☐	☐	
Interpret oceanographic information	☐	☐	
Create tactical recommendations			
Recommendations for movements, tactics and activities	☐	☐	
Recommendations for equipment usage and restrictions	☐	☐	
Ranges for detection and counter-detection	☐	☐	
Towed array speed	☐	☐	
Noise monitoring plan	☐	☐	
Create operational planning document	☐	☐	
ADDITIONAL COMMENTS			
STRENGTHS:	**AREAS FOR DEVELOPMENT:**		

ITEM	PASS	FAIL	COMMENTS
THE TRAINEE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			

OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

ENABLING CHECK 001.02 CONDUCT UWW BRIEF TO COMMAND

EC Time: 300 mins (total) + 50 min debrief (total)

ASSESSMENT ACTIVITY: Scenario continues from EC 001.01.

Trainees will use operational planning document and recommendations created in EC 001.01 to produce a Command Briefing for proposed operational activities to meet specific mission (from EC 001.01). Trainees will be provided with briefing template and EC checklist.

Operational planning document, including tactical recommendations, from EC 001.01 will be translated into a briefing that will provide Command with required information and recommendations to meet mission. Briefings must cover all required information, but only relevant content should be briefed, including all sections described in the EC checklist. (200 minutes, 4 periods)

Trainees will present the briefing to Command (guest assessor), with other trainees as part of the audience, instructor will use EC Checklist to assess overall briefing for content and delivery. Command will request justifications, clarifications and pose questions on brief and recommendations as appropriate. Command will also provide additional feedback based on brief received and delivery.

Immediate feedback will be provided by Comd based on material briefed and delivery. Additional feedback from the instructor will be provided in a group setting using the EC checklist and providing detailed feedback on content and delivery. (50 minutes, 1 period).

Pass/fail criteria: Trainees must achieve 5/5 SATs on the EC checklist.

ASSESSMENT RESOURCES:

- Operational planning document from EC 001.01
- Guest assessor (preferably current post-XO or above Submarine NWO)
- UWW briefing template
- SECRET computer with ability to display briefing slides

IMPACT OF FAILURE: Remedial Training / Re-test

INVIGILATOR NOTES:

- Individual EC feedback from instructor should be provided in a group setting unless trainee requests to have this individually.
- Command (guest assessor) should be sourced in advance of course and reconfirmed prior to this assessment.

1. PASS / FAIL CRITERIA: 5/5 SATs.

2. IMPACT OF FAILURE: Remedial training, re-test.

Underwater Controller 1 (AILE) –EC 001.02 Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

ITEM	PASS	FAIL	COMMENTS
UWW Brief to Command			
Briefing content meet requirements • Introduction • Classification level • References • Oceanography • Threat platforms • Recommendations	☐	☐	
Display confidence and professionalism • Eye contact • Body language and posture • Confidence and calm presence • Tone, language, delivery • Clear, concise, brief • Prepared and knowledgeable	☐	☐	
Provide appropriate recommendations • Based on available data; • Aligned with tactics and best practices • Relevant to capacities and capabilities • Meets aim of mission and Comd Intent	☐	☐	
Provide justification for recommendations • Explain reasoning • Answer questions with facts • Appropriate tone, language, and delivery	☐	☐	
Display critical thinking • Logical thinking evident • Reply to questions based on available data and situational awareness	☐	☐	
Briefing content meet requirements • Introduction • Classification level • References • Oceanography • Threat platforms • Recommendations	☐	☐	

ITEM	PASS	FAIL	COMMENTS
ADDITIONAL COMMENTS			
STRENGTHS:	**AREAS FOR DEVELOPMENT:**		
THE TRAINEE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			

OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

PERFORMANCE CHECK 001 PLAN SONAR OPERATIONAL ACTIVITIES

PC Time: 150 mins (total).

ASSESSMENT ACTIVITY: Injected in scenario for EO 002.01/EO 002.02 during trainer phase.

UC1 will be called to the watch room and given a development of new mission requirement (BARNSTORM), via briefing from OOW.
The trainee will be required to analyze the current situation, assess the requirements for sonar activities to support the mission, and make recommendations to the OOW.

The trainee, at plot, will verify the tactical picture with the OOW, confirm mission requirements, ask any pertinent questions from the OOW (who, what, where, when, why, how), noting that there is likely to be several unknown pieces of information at this time.
(5 minutes)

The trainee will be given 15 minutes to gather information and brief Command.

The trainee will then proceed to the sound room to brief the sonar team on mission requirements and current unknowns. The trainee will task the sound room to optimize the sonar settings for the new mission, research any available data in the publications onboard, and verify the oceanographic conditions.

Expected analysis of the UC1 during this period will include:
- assess UWW threat information;
- identify capacities and limitations of platforms;
- interpret oceanographic data and determine their effects on acoustic operations;
- calculate acoustic sensor range information;
- consider noise monitoring plan amendments; and
- apply water space management;

(15 minutes)

Using provided information from the sound room, publications, and current oceanographic conditions, the trainee will brief Command with recommendations based on available information and highlight known and unknown information for threat and oceanographic conditions.
(5 minutes)

Feedback will be given immediately by Command on the recommendations briefed. Detailed feedback will be provided by the instructor following completion of the scenario using the PC checklist, during trainee down-time as observer of other runs

Pass/fail criteria: Trainee must achieve 14/16 SATs on the PC checklist to pass this assessment.

ASSESSMENT RESOURCES:
- BARNSTOM scenario inject;
- VCSCTT / TWS

IMPACT OF FAILURE: IAW NTORD 4500-54

INVIGILATOR NOTES:

- OOW briefing should include recommendations for which the UC1 should recommend against and provide justification for this counter-recommendation;
- This assessment will take place during other scenario runs for EO 002.02 / 002.03.
- This assessment will not cover the amplification and changes to the situation following the initial brief.

1. **PASS / FAIL CRITERIA:** 14/16 SATs.

2. **IMPACT OF FAILURE:** IAW NTORD 4500-54.

Underwater Controller 1 (AILE) – PC 001 Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

ITEM	**PASS**	**FAIL**	**COMMENTS**
Sonar Operational Planning			
Review given mission requirements and type	☐	☐	
Collect data to support tactical decision making	☐	☐	
Identify and request additional data	☐	☐	
Analyze data for impact on mission			
Assess UWW threat information	☐	☐	
Identify capacities and limitations of platforms	☐	☐	
Interpret oceanographic information	☐	☐	
Create tactical recommendations			
Recommendations for movements, tactics and activities	☐	☐	
Recommendations for equipment usage and restrictions	☐	☐	
Ranges for detection and counter-detection	☐	☐	
Towed array speed	☐	☐	
Noise monitoring plan	☐	☐	
Conduct UWW Brief to Command			
Briefing content meet requirements for mission	☐	☐	
Display confidence and professionalism	☐	☐	
Provide appropriate recommendations	☐	☐	
Provide justification for recommendations	☐	☐	

ITEM	PASS	FAIL	COMMENTS
Display critical thinking	☐	☐	

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE TRAINEE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

ENABLING CHECK 002.01 LEAD SOUND ROOM IN SUPPORT OF OPERATIONS

EC Time: 300 mins (total) + 50 min debrief (total)

ASSESSMENT ACTIVITY: Trainee UC1 will be provided the pre-mission brief from Command (acted by instructor), including mission, situation, contacts/threats, and ROE. (10 minutes)

Trainee will direct the set up and configuration required for departure, including confirmation that equipment is functioning as required, operational logs and records are prepared as required.

The trainee will brief sonar watch (or SWL) on situation, including required actions to maintain ordered quiet state. Upon confirmation from Command that sonar watch is closed up, the trainee will leave the trainer and be on standby in briefing room.

Trainee on standby will be recalled to the trainer as situation dictates UC1 presence/action.

Potential situations include (A minimum of 3 situations should be covered during the PC, and a maximum of 6):

- Whenever Command or watch is in doubt;
- Return to periscope depth;
- Equipment failures;
- Personnel issues;
- Abrupt-in contacts;
- Fishing vessels;
- Close-quarter contacts;
- Contacts of interest (COI);
- Submarines;
- Torpedoes;
- Active transmissions;
- Aircraft on top;
- Warships;
- Self-noise monitoring;
- Transient;
- Action stations;
- Weapons firing;
- Attack teams; and
- EOPs.;

Upon return to trainer, UC1 will receive briefing from OOW and Sonar Watch Leader on the situation, take charge of the sound room, and will be required to confirm tactical picture and information, analyze data, and assess the situation. Based on this assessment, the UC1 will recommend actions to Command to achieve mission and priorities. The UC1 will be expected to justify and amplify recommendations as required by Command.

Note: this scenario will not include towed array operations.

(40 minutes per trainee)

Trainees will be advised immediately following assessment if they have achieved the standard and then individual feedback will be provided using the PC checklist to the trainees following completion of all PCs (50 minutes total).

Pass/fail criteria: Trainees must achieve 5/5 SATs on EC checklist.

ASSESSMENT RESOURCES: • VCSCTT / TWS **IMPACT OF FAILURE:** Remedial Training / Re-test

INVIGILATOR NOTES:

- Requests for NWO, Trainer Staff, and by-standers / extra support personnel should be submitted to Submarine Division Training Coordinator prior to course start.
- Given requirement to have supplemental staff to act as supporting personnel, it should be noted that the overall focus should be on UC1 conduct, not on the quality of sound room personnel's actions.

1. PASS / FAIL CRITERIA: 5/5 SATs.
2. IMPACT OF FAILURE: Remedial training, re-test.

Underwater Controller 1 (AILE) – EC 002.01 Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

ITEM	**PASS**	**FAIL**	**COMMENTS**
Lead sound room in support of operations			
Optimize ship's acoustic signature by making recommendations for movements to optimize tactical posture	☐	☐	
Oversee tactical picture	☐	☐	
Provide recommendations to command (Situation, Assessment, Recommendation)	☐	☐	
Monitor operation of all submarine sonar equipment	☐	☐	
Direct maintenance of operational logs and records	☐	☐	

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE TRAINEE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

ENABLING CHECK 002.02 LEAD SOUND ROOM IN SUPPORT OF TOWED ARRAY OPERATIONS

EC Time: 300 mins (total) + 50 min debrief (total)

ASSESSMENT ACTIVITY: Trainee UC1 will be provided the pre-mission brief from Command (acted by instructor), including mission, situation, contacts/threats, and ROE. (10 minutes)

Trainee will direct the set up and configuration required for departure, including confirmation that equipment is functioning as required, operational logs and records are prepared as required.

The trainee will brief sonar watch (or SWL) on situation, including required actions to maintain ordered quiet state. Upon confirmation from Command that sonar watch is closed up, the trainee will leave the trainer and be on standby in briefing room.

Trainee on standby will be recalled to the trainer as situation dictates UC1 presence/action.

Potential situations include (A minimum of 3 situations should be covered during the PC, and a maximum of 6):

- Whenever Command or watch is in doubt;
- Return to periscope depth;
- Equipment failures;
- Personnel issues;
- Abrupt-in contacts;
- Fishing vessels;
- Close-quarter contacts;
- Contacts of interest (COI);
- Submarines;
- Torpedoes;
- Active transmissions;
- Aircraft on top;
- Warships;
- Self-noise monitoring;
- Transient;
- Action stations;
- Weapons firing;
- Attack teams; and
- EOPs.

Upon return to trainer, UC1 will receive briefing from OOW and Sonar Watch Leader on the situation, take charge of the sound room, and will be required to confirm tactical picture and information, analyze data, and assess the situation. Based on this assessment, the UC1 will recommend actions to Command to achieve mission and priorities. The UC1 will be expected to justify and amplify recommendations as required by Command.

Note: this scenario must include towed array operations.

(40 minutes per trainee)

Trainees will be advised immediately following assessment if they have achieved the standard and then individual feedback will be provided using the PC checklist to the trainees following completion of all PCs (50 minutes total).

Pass/fail criteria: Trainees must achieve 6/6 SATs on PC checklist. **ASSESSMENT RESOURCES:** • VCSCTT / TWS **IMPACT OF FAILURE:** Remedial Training / Re-test

INVIGILATOR NOTES:

- Requests for NWO, Trainer Staff, and by-standers / extra support personnel should be submitted to Submarine Division Training Coordinator prior to course start.
- Given requirement to have supplemental staff to act as supporting personnel, it should be noted that the overall focus should be on UC1 conduct, not on the quality of sound room personnel's actions.

1. **PASS / FAIL CRITERIA:** 6/6 SATs.

2. **IMPACT OF FAILURE:** Remedial training, re-test.

Underwater Controller 1 (AILE) – EC 002.02 Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

ITEM	**PASS**	**FAIL**	**COMMENTS**
Lead sound room in support of towed array operations			
Optimize ship's acoustic signature by making recommendations for movements to optimize tactical posture	☐	☐	
Oversee tactical picture	☐	☐	
Direct towed array system operations	☐	☐	
Provide recommendations to command (Situation, Assessment, Recommendation)	☐	☐	
Monitor operation of all submarine sonar equipment	☐	☐	
Direct maintenance of operational logs and records	☐	☐	

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE TRAINEE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:

ITEM		PASS	FAIL	COMMENTS
Trainee	Name:	Date:	Signature:	

PERFORMANCE CHECK 002 LEAD SOUND ROOM IN SUPPORT OF OPERATIONS

PC Time: 600 mins (total) + 50 min debrief (total)

ASSESSMENT ACTIVITY: Trainee UC1 will be provided the pre-mission brief from Command (acted by instructor), including mission, situation, contacts/threats, and ROE. (10 minutes)

Trainee will direct the set up and configuration required for departure, including confirmation that equipment is functioning as required, operational logs and records are prepared as required.

The trainee will brief sonar watch (or SWL) on situation, including required actions to maintain ordered quiet state. Upon confirmation from Command that sonar watch is closed up, the trainee will leave the trainer and be on standby in briefing room.

Trainee on standby will be recalled to the trainer as situation dictates UC1 presence/action.

Potential situations include (A minimum of 3 situations should be covered during the PC, and a maximum of 6):

- Whenever Command or watch is in doubt;
- Return to periscope depth;
- Equipment failures;
- Personnel issues;
- Abrupt-in contacts;
- Fishing vessels;
- Close-quarter contacts;
- Contacts of interest (COI);
- Submarines;
- Torpedoes;
- Active transmissions;
- Aircraft on top;
- Warships;
- Self-noise monitoring;
- Transient;
- Action stations;
- Weapons firing;
- Attack teams; and
- EOPs.

Upon return to trainer, UC1 will receive briefing from OOW and Sonar Watch Leader on the situation, take charge of the sound room, and will be required to confirm tactical picture and information, analyze data, and assess the situation. Based on this assessment, the UC1 will recommend actions to Command to achieve mission and priorities. The UC1 will be expected to justify and amplify recommendations as required by Command.

Note: this scenario must include towed array operations.

(90 minutes per trainee)

Trainees will be advised immediately following assessment if they have achieved the standard and then individual feedback will be provided using the PC checklist to the trainees following completion of all PCs (50 minutes total). **Pass/fail criteria**: Trainees must achieve 6/6 SATs on PC checklist. **ASSESSMENT RESOURCES:** • VCSCTT / TWS **IMPACT OF FAILURE:** IAW NTORD 4500-54

INVIGILATOR NOTES:

- Requests for NWO, Trainer Staff, and by-standers / extra support personnel should be submitted to Submarine Division Training Coordinator prior to course start.
- Given requirement to have supplemental staff to act as supporting personnel, it should be noted that the overall focus should be on UC1 conduct, not on the quality of sound room personnel's actions.

1. PASS / FAIL CRITERIA: 6/6 SATs.

2. IMPACT OF FAILURE: IAW NTORD 4500-54.

Underwater Controller 1 (AILE) – PC 002 Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

ITEM	**PASS**	**FAIL**	**COMMENTS**
Lead sound room in support of towed array operations			
Optimize ship's acoustic signature by making recommendations for movements to optimize tactical posture	☐	☐	
Oversee tactical picture	☐	☐	
Direct towed array system operations	☐	☐	
Provide recommendations to command (Situation, Assessment, Recommendation)	☐	☐	
Monitor operation of all submarine sonar equipment	☐	☐	
Direct maintenance of operational logs and records	☐	☐	
ADDITIONAL COMMENTS			
STRENGTHS:	**AREAS FOR DEVELOPMENT:**		

ITEM	PASS	FAIL	COMMENTS
THE TRAINEE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			

OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

ENABLING CHECK 003.01 DEVELOP SONAR TEAM IN SUPPORT OF OPERATIONS

EC Time: 100 mins (total)

ASSESSMENT ACTIVITY: Trainees will be provided with upcoming mission scenario for which they will need to lead skill development training for the Sonar Operators. Instructor will provide trainees with a template training development plan.

Trainees will be required to plan a training activity including the following:

- Identify training requirements based on provided mission (i.e. which skills require development training)
- Select appropriate training type (individual or team training) to meet developmental requirements for mission
- Develop a scenario that will be used to conduct this training
- Identify required training supports to conduct training, such as:
  - Equipment required
  - Facilities/location required
  - Personnel required to participate and support
  - Data required
  - Supporting material, such as answer key, AIGs, publications, references
- Identify assessment factors
- Create feedback plan

Trainees will create the plan on their own time and submit to instructor for instructor review and assessment.

Trainees will be provided one period (50 minutes) with the instructor to ask questions, seek clarifications, and discuss planning options. After this, trainees will be expected to conduct on own time and draft the plan.

The group will reconvene to discuss draft plans (50 minutes), seek input from fellow trainees and instructor based on their draft plans, then trainees will have further opportunity to finalize the plan before submission.

(100 minutes of class time, remainder self-paced)

Group and individual feedback will be provided during second course period, instructor will answer questions and provide feedback on draft plans and trainees will have further opportunity to collaborate.

Individual feedback will be provided on submitted assignments with EC checklist from instructor

Pass/fail criteria: Trainee must attain 5/6 SATs to pass this assessment.

ASSESSMENT RESOURCES:

- Training activity template
- Scenario
- Classroom

IMPACT OF FAILURE: Remedial Training / Re-test

INVIGILATOR NOTES:

Underwater Controller 1 (AILE) – EC 003.01 Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

1. **PASS / FAIL CRITERIA:** 5/6 SATs.

2. **IMPACT OF FAILURE:** Remedial training, re-test.

ITEM	PASS	FAIL	COMMENTS
Develop Sonar team in support of operations			
Identify training requirements based on scenario	☐	☐	
Select appropriate training type (team or individual)	☐	☐	
Develop scenario for training	☐	☐	
Identify required training supports	☐	☐	
Identify assessment factors	☐	☐	
Create feedback plan	☐	☐	

ADDITIONAL COMMENTS	
STRENGTHS:	**AREAS FOR DEVELOPMENT:**

THE TRAINEE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

MARPAC RDIMS #1387867
Chap 1 & 2 approved 01/12/2023
Chap 3 & 4 approved 18/01/2024

ANNEX J – ON JOB PERFORMANCE RECORD (OJPR)

Nil.

ANNEX K – TT PLAN

Nil.

ANNEX L – RECORD OF DECISIONS TPWB

1. This record summarizes the conduct of the TPWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

See QSWB Record of Decisions, Annex D.

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the TP is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted TP Change Request (TPCR), thus ensuring the QSP has the most relevant information.

Version:	**X.X**
Date:	**Who Initiated Change:**
Requirement for the Amendment:	
Description of Changes:	
Change Tracking number:	
Trg Mgr:	

ANNEX M – ACRONYMS AND ABBREVIATIONS

1. This annex is to be utilized only to identify uncommon abbreviations used within this QSP.

A	
ASW	Anti-Submarine Warfare
ATT	Actual Training Time
B	
BAT	Base Administrative Time
BSC	Basic Submarine Course
C	
CAF	Canadian Armed Forces
CFITES	Canadian Forces Individual Training and Education System
Cmdt	Commandant
CNTG	Commander Naval Training Group
CO	Commanding Officer
CSNI	Consolidated Secret Network Infrastructure
CSO	Captain's Standing Orders
D	
DAOD	Defence Administrative Orders and Directives
DSCDS	Defence Subject Classification and Disposition System
DND	Department of National Defence
DPGR	Director Personnel Generation Requirements
DWAN	Defence Wide Area Network
E	
EC	Enabling Check
EDI	Explain, Demonstrate, Imitate
EO	Enabling Objective
G	
GPNet	General Purpose Network
H	
HOD	Head of Department
I	
IAW	In Accordance With
ID	Identification
IMS	Information Management Standard
IS	Information Systems
IT	Information Technology
ITEM	Interactive Electronic Technical Manual
L	

LO	Learning Objective
LG	Lesson Guidance
M	
MITE	Military Individual Training and Education
MOSID	Military Occupational Structure Identification Code
N	
NCPM	Naval Combat Operating Procedures Manual
NQual	National Qualification
NFS	Naval Fleet School
NAVORD	Naval Order
NTG	Naval Training Group
NPTORD	Naval Personnel and Training Order
NWO	Naval Warfare Officer
O	
OJT	On-the-Job Training
OJPR	On-the-Job Performance Requirement
OOW	Officer of the Watch
P	
PAL	Passive Aural Listening
PC	Performance Check
PO	Performance Objective
PLAR	Prior Learning Assessment and Recognition
PT	Physical Training
Q	
QSP	Qualification Standard and Plan
QSPWB	Qualification Standard and Plan Writing Board
R	
R&Q	Rations & Quarters
RCN	Royal Canadian Navy
RSS	Request for Service Support
S	
SAR	Situation, Assessment, Recommendation
SAT	Satisfactory
SO	Standing Order
SOP	Standard Operating Procedure
SWL	Sonar Watch Leader
T	
TAT	Training Administrative Time
TA	Training Authority
TD	Temporary Duty
TE	Training Establishment
TEL	Technology Enabled Learning
TO	Training Officer
TPO	Training Petty Officer
TRB	Training Review Board

TS	Training Site
TTT	Training Travel Time
TWS	Tactical Weapons System
U	
UC1	Underwater Controller 1
UNCLAS	Unclassified
UNSAT	Unsatisfactory
UWW	Under Water Warfare
V	
VCSOP	Victoria Class Standard Operating Procedures
VCSSO	Victoria Class Ship’s Standing Orders
VCSCTT	Victoria Class Submarine Command Team Trainer
Vol	Volume